Universidad Internacional del Ecuador (UIDE)

Computer Science School

Network Communication Technology

Title: *Phase 2: Packet Tracer Network Prototype*

Professor: Steven Vinueza. M.Sc.

Team Members

Santiago Alejandro Lucina Mena

Pamela Stefania Toapanta Pila

Camila Isabel Paucar Guaranda

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1. UPDATED ARCHITECTURE

The design provided in Phase 1 laid the foundation for defining the company's final network with all of its services and configurations. For this initial prototype in Cisco Packet Tracer, the structure was simplified to clearly verify that addressing, basic connectivity, the DHCP service, and communication between sites work correctly:

- **Use of Cisco Routers in Simulation:** Cisco routers are used in Packet Tracer to validate routes, gateways, and the dynamic delivery of IP addresses before moving to the final OpenWrt devices.
- **Simplified Switch Connectivity:** One main switch per site is used, operating in simple access mode. This makes it possible to verify the physical connectivity of the devices and MAC address learning without adding complex redundancy configurations in this first release.
- **Reserved Segmentation Plan:** To avoid overloading the initial prototype, the test segments are activated directly on the local links. The subnets needed for the other work groups are calculated and reserved within the addressing plan, ready to be assigned to their respective VLANs in the next stage.
- **Maintaining the WAN Link:** The 192.168.100.0/24 network originally established to connect both sites through their edge interfaces is kept.

Implemented and Planned Elements

Table 1

Implemented and Planned Elements

COMPONENT	IMPLEMENTED ELEMENTS	PLANNED ELEMENTS
PERIMETER ROUTING	Cisco routers with active interfaces and an inter-site WAN link on 192.168.100.0/24. Connectivity verified between both sites.	Deployment on OpenWrt routers with full OSPFv2 and control of routes to the Internet.
LOCAL SWITCHING	Cisco 2960 switches with access ports for computers and servers. MAC address table verification.	Addition of a second switch per site, redundant links with EtherChannel, and loop prevention with RSTP.
NETWORK SEGMENTATION	VLSM subnets calculated from the 10.10.0.0/16 corporate block, tested on the active segments.	Formal VLAN configuration, 802.1Q trunk links, and inter-VLAN routing.
NETWORK SERVICES	DHCP server active on the routers to assign IP addresses to the test computers, reserving the IPs of servers and gateways.	Independent DHCP servers for each VLAN on the perimeter router, firewall rules, and network address translation (NAT).
SOFTWARE SERVICES	Simulated client and test servers to verify the sending and receiving of packets on the network.	Connection of the Go messaging system, the FastAPI authentication service, and the database.

Note. Created by the author from the Cisco Packet Tracer simulation environment. It shows the distribution of active interfaces and physical links between sites.

Prototype Connections and Interfaces

Site A

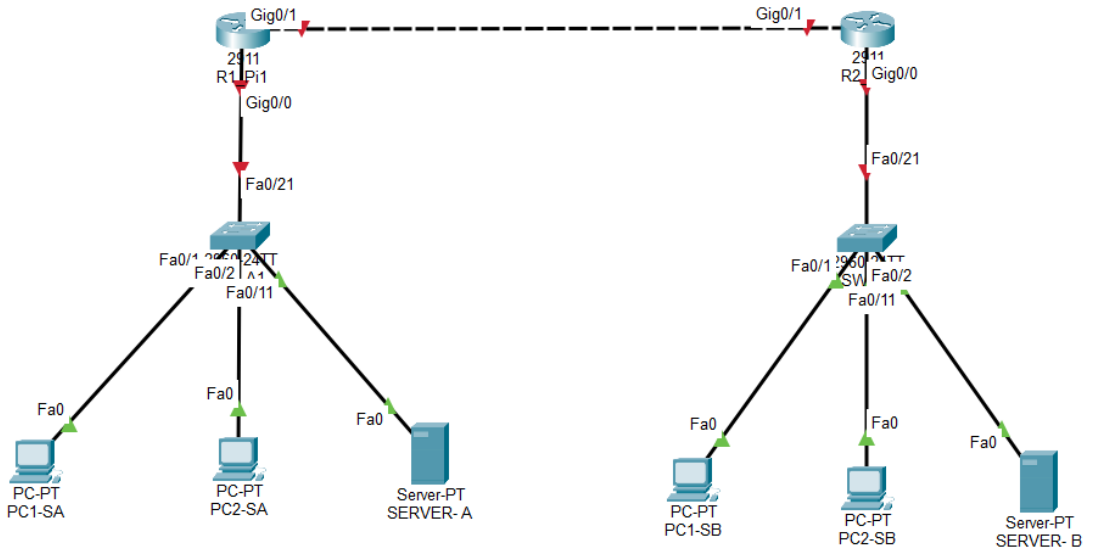
- **Edge Router (R1_Pi1):**
  - **GigabitEthernet0/1:** Connection to the inter-site WAN link with IP 192.168.100.10/24.
  - **GigabitEthernet0/0:** Connection to the local LAN with IP 10.10.0.1/26.
- **Access Switch (SW-A1):**
  - **FastEthernet0/21:** Uplink connected to R1_Pi1.
  - **FastEthernet0/1**: Access connection for workstation PC1-SA.
  - **FastEthernet0/2:** Access connection for workstation PC2-SA.
  - **FastEthernet0/11:** Access connection for the local server SERVER-A.

Site B

- **Edge Router (R2_Pi2):**
  - **GigabitEthernet0/1:** Connection to the inter-site WAN link with IP 192.168.100.11/24.
  - **GigabitEthernet0/0:** Connection to the local LAN with IP 10.10.1.1/26.
- **Access Switch (SW-B1):**
  - **FastEthernet0/21:** Uplink connected to R2_Pi2.
  - **FastEthernet0/1:** Access connection for workstation PC1-SB.
  - **FastEthernet0/2:** Access connection for workstation PC2-SB.
  - **FastEthernet0/11:** Access connection for the local server SERVER-B

Figure 1

Prototype Connections in Cisco Packet Tracer



Note. Created by the author in Cisco Packet Tracer. It shows the distribution of active interfaces and physical links between sites.

2. VLSM ADDRESSING

To organize the network efficiently and avoid wasting IP addresses, the VLSM method was used starting from the main block 10.10.0.0/16.

To keep a clear order and allow routes to be summarized easily, the address space was divided into two main groups:

- **Site A:** Assigned the base space 10.10.0.0/24.
- **Site B:** Assigned the base space 10.10.1.0/24.

VLAN 10 - Users (30 devices per site)

- With 6 host bits, 62 usable addresses are obtained ($2^6 - 2 = 62$).

- The /26 prefix (mask 255.255.255.192) is used, occupying a block of 64 addresses.

VLAN 30 - Guests (15 devices per site)

- With 5 host bits, 30 usable addresses are obtained ($2^5 - 2 = 30$).
- The /27 prefix (mask 255.255.255.224) is used, occupying a block of 32 addresses.

VLAN 20 - Servers (4 devices per site)

- With 4 host bits, 14 usable addresses are obtained ($2^4 - 2 = 14$).
- The /28 prefix (mask 255.255.255.240) is used, occupying a block of 16 addresses.

VLAN 99 - Management (4 devices per site)

- With 4 host bits, 14 usable addresses are obtained ($2^4 - 2 = 14$).
- The /28 prefix (mask 255.255.255.240) is used, occupying a block of 16 addresses.

Inter-site WAN link:

- Uses the shared network 192.168.100.0/24 defined previously.

Table 2

VLSM Addressing Derived from the 10.10.0.0/16 Block

SITE AND SEGMENT	PURPOSE	HOSTS REQUIRED	NETWORK	PREFIX	SUBNET MASK	USABLE IP RANGE	BROADCAST	GATEWAY	ASSIGNMENT
SITE A - VLAN 10	Users and Operations	30	10.10.0.0	/26	255.255.255.192	10.10.0.1 to 10.10.0.62	10.10.0.63	10.10.0.1	DHCP

SITE A - VLAN 30	Guests	15	10.10.0.64	/27	255.255.255.224	10.10.0.65 to 10.10.0.94	10.10.0.95	10.10.0.65	DHCP
SITE A - VLAN 20	Internal servers	4	10.10.0.96	/28	255.255.255.240	10.10.0.97 to 10.10.0.110	10.10.0.111	10.10.0.97	Static
SITE A - VLAN 99	Management	4	10.10.0.112	/28	255.255.255.240	10.10.0.113 to 10.10.0.126	10.10.0.127	10.10.0.113	Static
SITE B - VLAN 10	Users and Operations	30	10.10.1.0	/26	255.255.255.192	10.10.1.1 to 10.10.1.62	10.10.1.63	10.10.1.1	DHCP
SITE B - VLAN 30	Guests	15	10.10.1.64	/27	255.255.255.224	10.10.1.65 to 10.10.1.94	10.10.1.95	10.10.1.65	DHCP
SITE B - VLAN 20	Local servers	4	10.10.1.96	/28	255.255.255.240	10.10.1.97 to 10.10.1.110	10.10.1.111	10.10.1.97	Static
SITE B - VLAN 99	Management	4	10.10.1.112	/28	255.255.255.240	10.10.1.113 to 10.10.1.126	10.10.1.127	10.10.1.113	Static
WAN LINK	Connection between sites	2	192.168.100.0	/24	255.255.255.0	192.168.100.1 to 192.168.100.254	192.168.100.255	Not applicable	Static

Note. Created by the author. The Guests and Management subnets are calculated and reserved for implementation in the advanced switching stage.

IP Address Assignment

In the current Packet Tracer prototype, addresses are assigned as follows:

Site A

- **Router R1_Pi1 (Interface Gi0/0 - Local network):** 10.10.0.1/26
- **Router R1_Pi1 (Interface Gi0/1 - WAN link):** 192.168.100.10/24
- **Switch SW-A1 (Temporary management):** 10.10.0.2/26
- **Server SERVER-A:** 10.10.0.3/26
- **Computers PC1-SA and PC2-SA:** Receive their address by DHCP in the range 10.10.0.10 to 10.10.0.62.

Site B

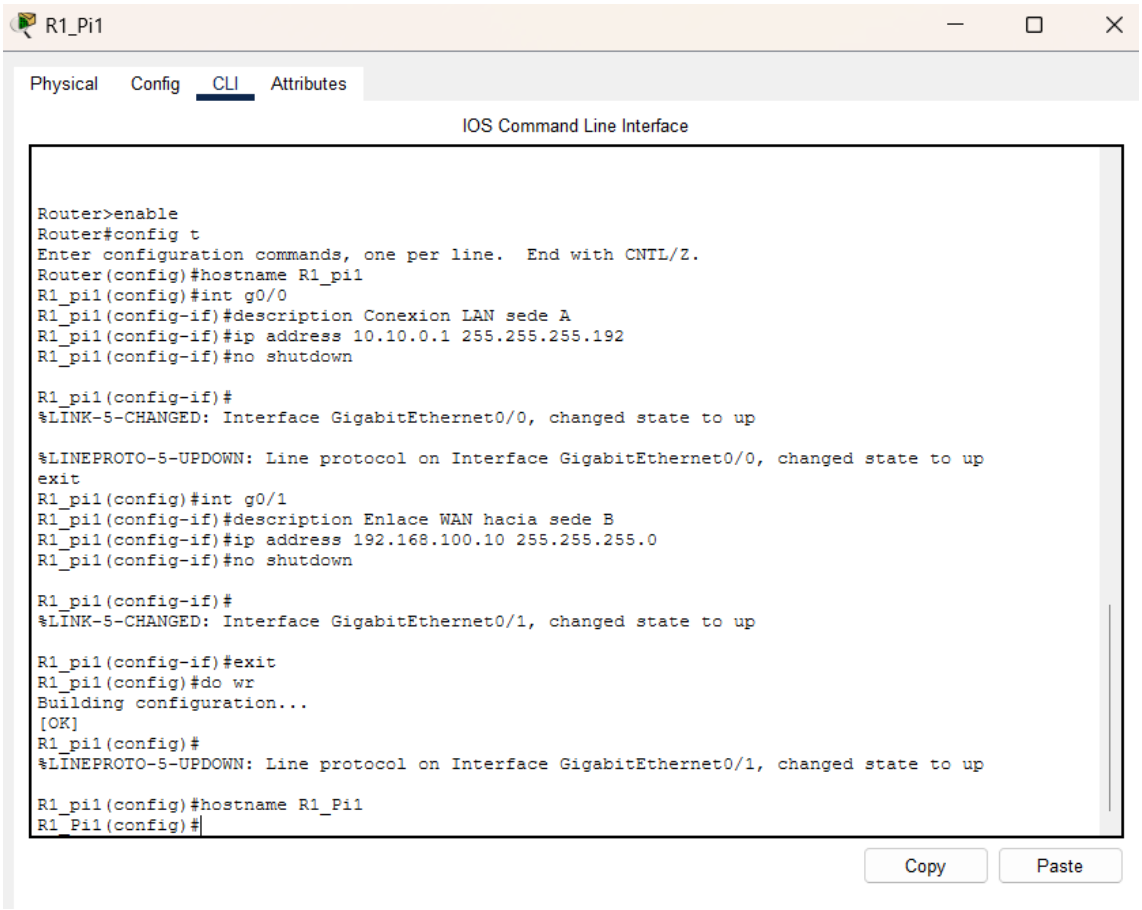
- **Router R2_Pi2 (Interface Gi0/0 - Local network):** 10.10.1.1/26
- **Router R2_Pi2 (Interface Gi0/1 - WAN link):** 192.168.100.11/24
- **Switch SW-B1 (Temporary management):** 10.10.1.2/26
- **Server SERVER-B:** 10.10.1.3/26
- **Computers PC1-SB and PC2-SB:** Receive their address by DHCP in the range 10.10.1.10 to 10.10.1.62.

3. BASIC DEVICE CONFIGURATION

This section documents the basic configuration process applied to the routing and switching devices in the topology.

Figure 2

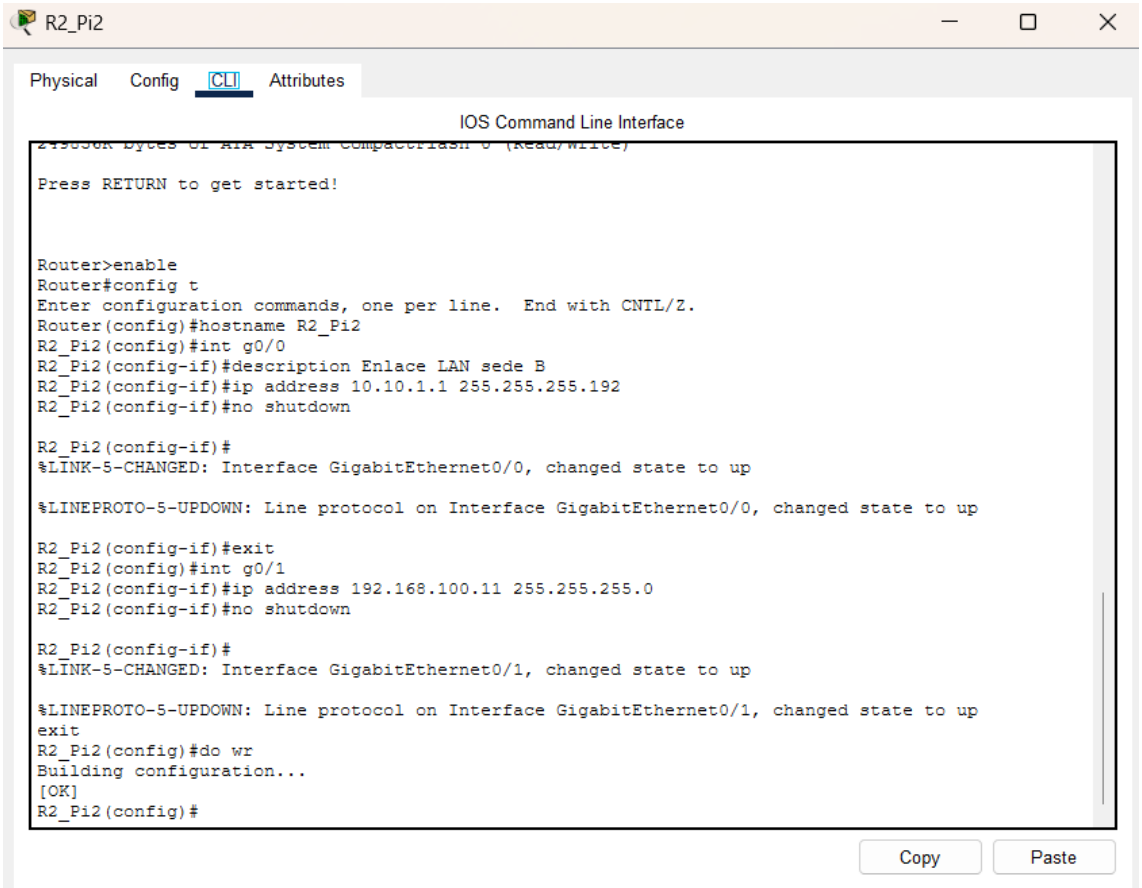
Initial Configuration of Router R1_Pi1 in Cisco Packet Tracer



Note. Created by the author from the Cisco Packet Tracer command-line interface.

Figure 3

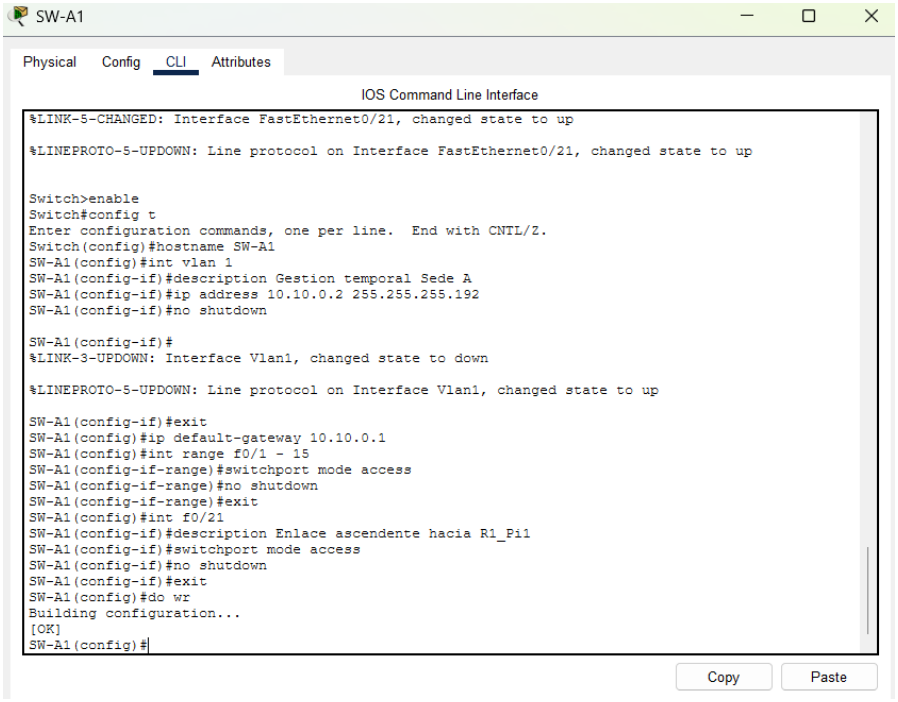
Initial Configuration of Router R2_Pi2 in Cisco Packet Tracer



Note. Created by the author from the Cisco Packet Tracer command-line interface.

Figure 4

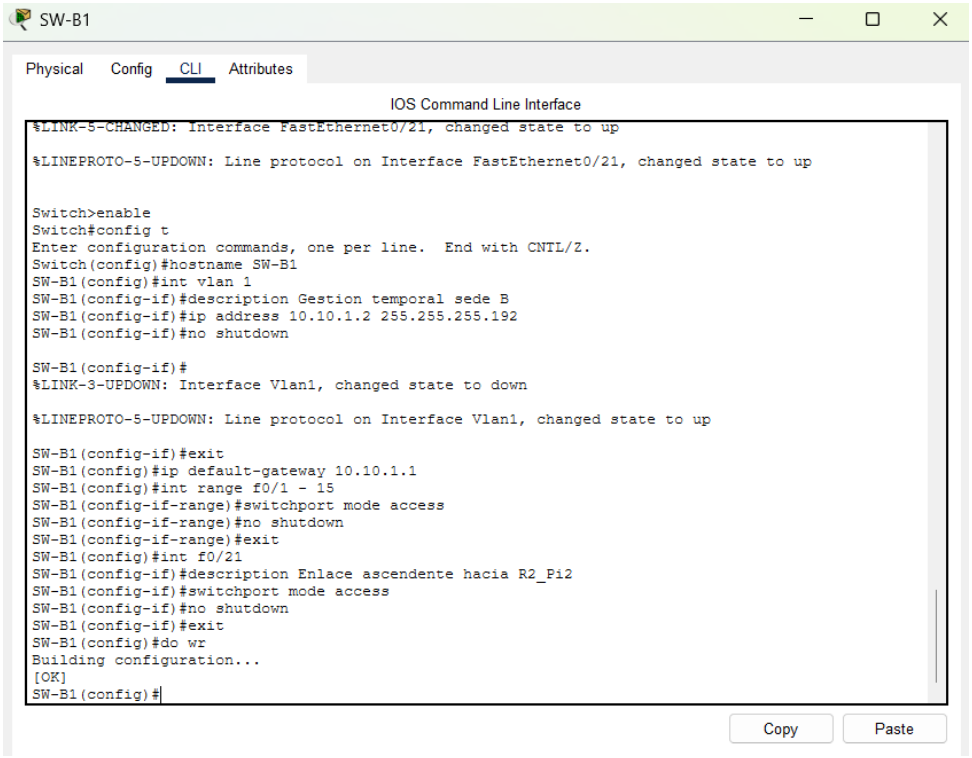
Configuration of Switch SW-A1 in Cisco Packet Tracer



Note. Created by the author from the Cisco Packet Tracer command-line interface.

Figure 5

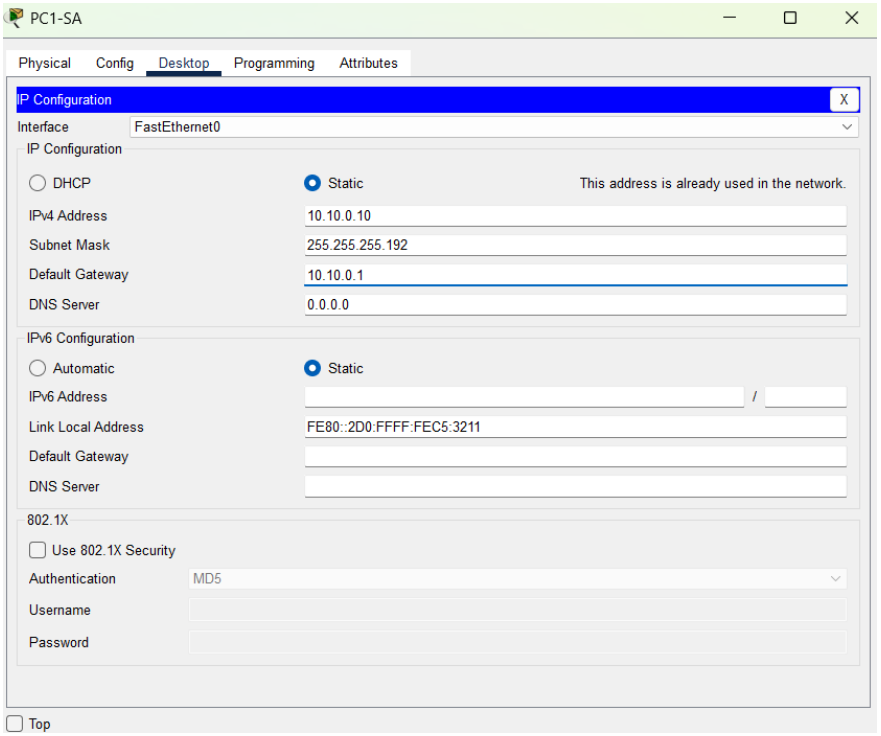
Configuration of Switch SW-B1 in Cisco Packet Tracer



Note. Created by the author from the Cisco Packet Tracer command-line interface (CLI).

Figure 6

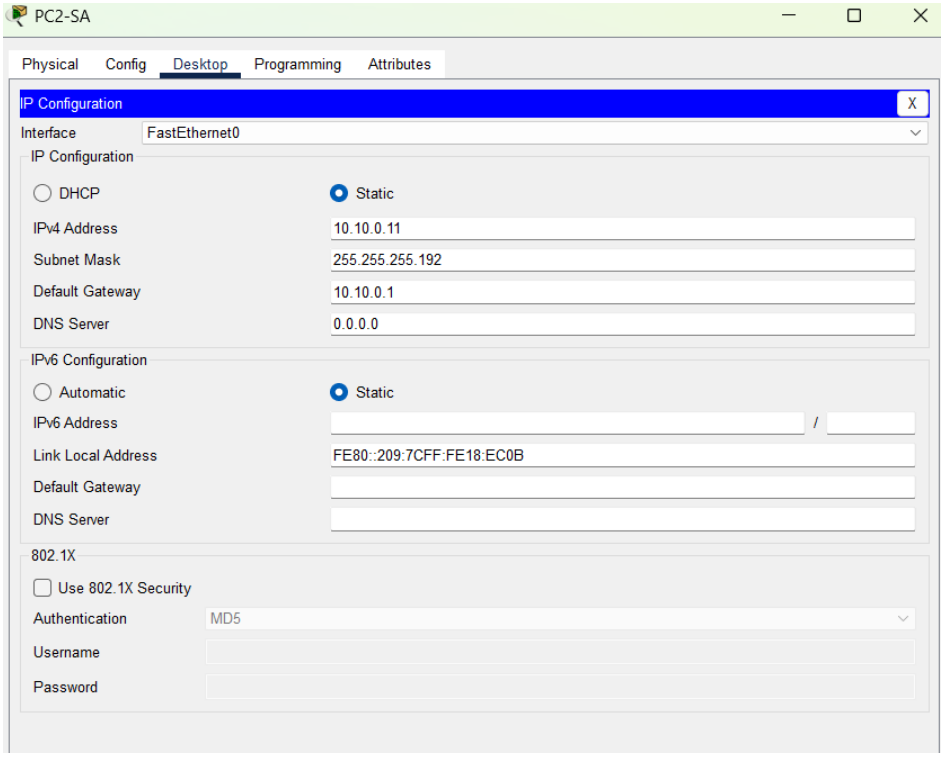
Preliminary Static IP Configuration on Workstation PC1-SA



Note. Created by the author from the Cisco Packet Tracer configuration panel.

Figura 7

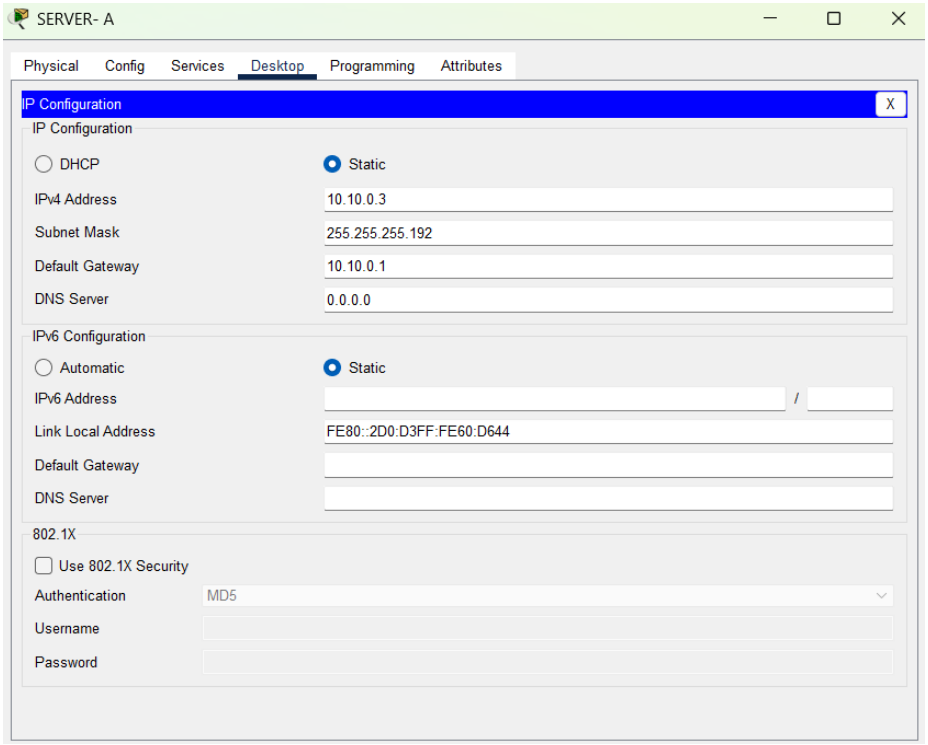
Preliminary Static IP Configuration on Workstation PC2-SA



Note. Created by the author from the Cisco Packet Tracer configuration panel.

Figura 8

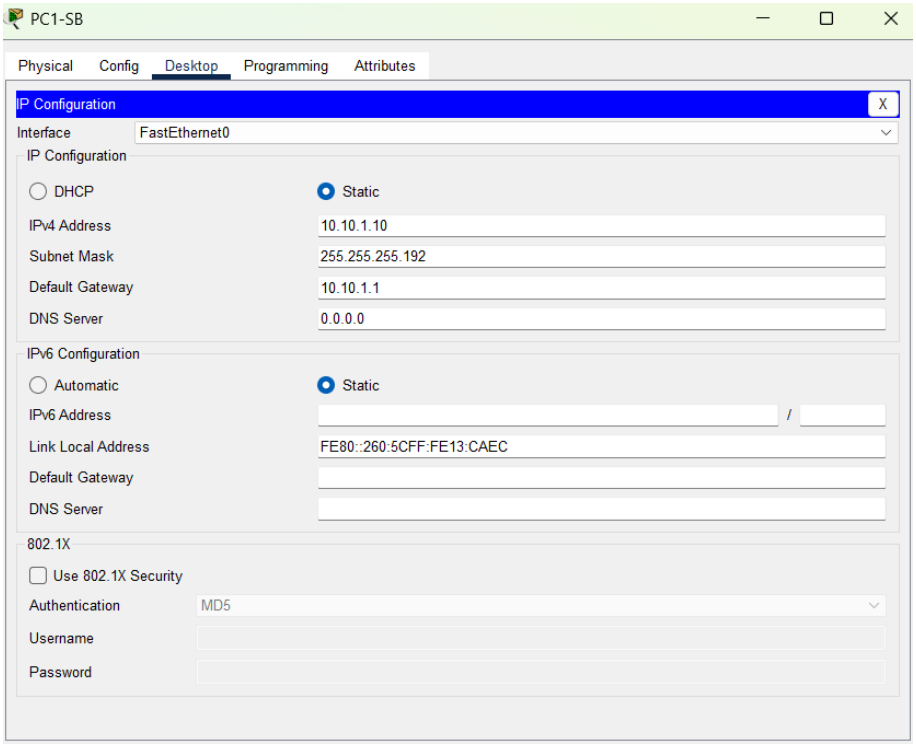
Static IP Configuration on Corporate Server SERVER-A



Note. Created by the author from the Cisco Packet Tracer configuration panel.

Figura 9

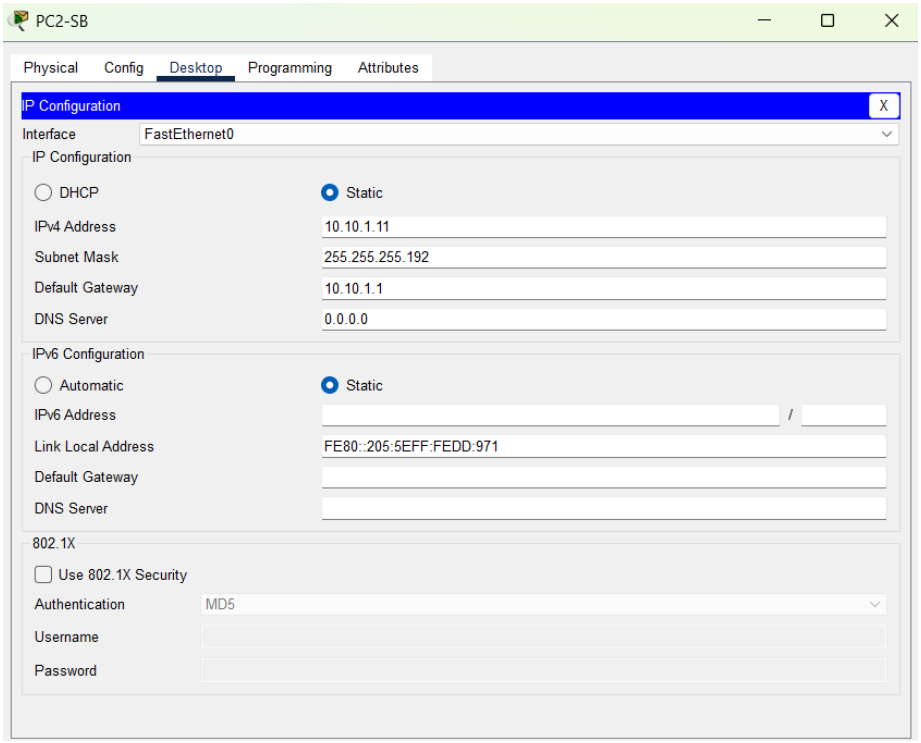
Preliminary Static IP Configuration on Workstation PC1-SB



Note. Created by the author from the Cisco Packet Tracer configuration panel.

Figura 10

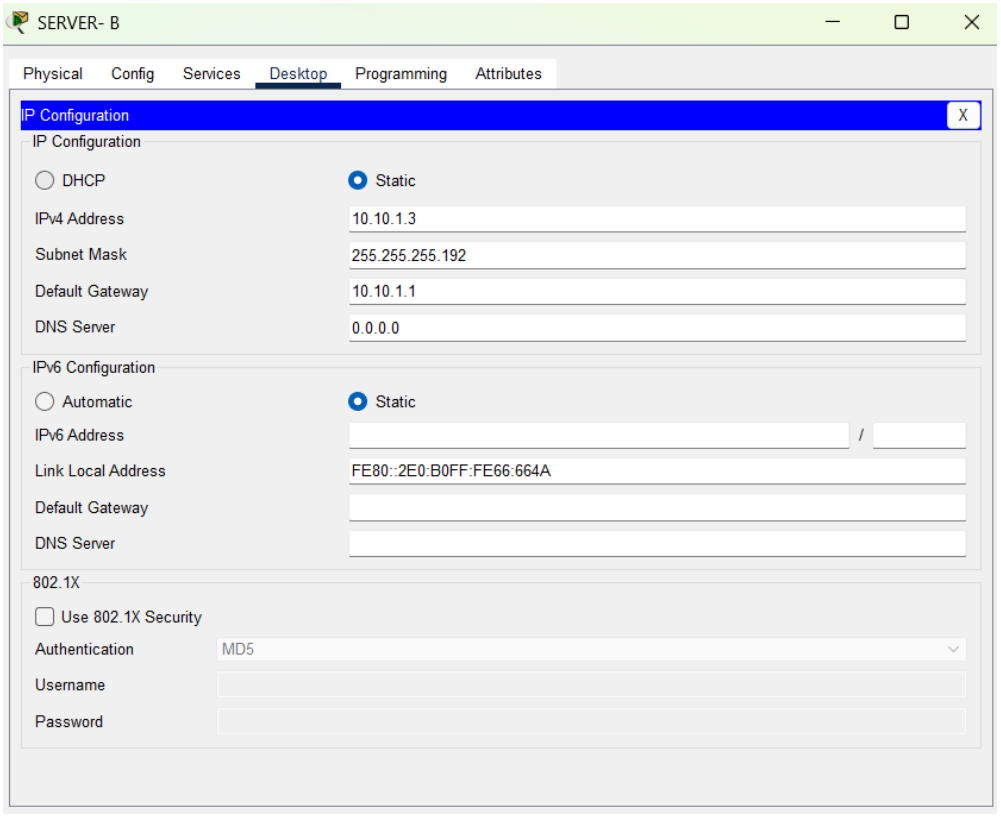
Preliminary Static IP Configuration on Workstation PC2-SB



Note. Created by the author from the Cisco Packet Tracer configuration panel.

Figura 11

Static IP Configuration on Corporate Server SERVER-B



Note. Created by the author from the Cisco Packet Tracer configuration panel..

Verification of Active Interfaces

Interface status was verified on both routers using the *do show ip interface brief* command. This check confirms that the physical and data link layers are operational on the active links.

Figure 12

Verification of Interface Status on R1_Pi1 and R2_Pi2



Note. Created by the author from running *do show ip interface brief* on the edge routers.

As shown in Figure 12, interfaces GigabitEthernet0/0 and GigabitEthernet0/1 on both routers show an operational up/up state. The remaining interfaces stay administratively disabled as an infrastructure security measure.

MAC Address Table Inspection and Explanation

While transmitting test frames between the end devices, the Layer 2 forwarding table on SW-A1 and SW-B1 was inspected using the *do show mac address-table* command to confirm that the physical addresses were registered.

Figure 13

Dynamic MAC Address Table Learned on SW-A1 and SW-B1

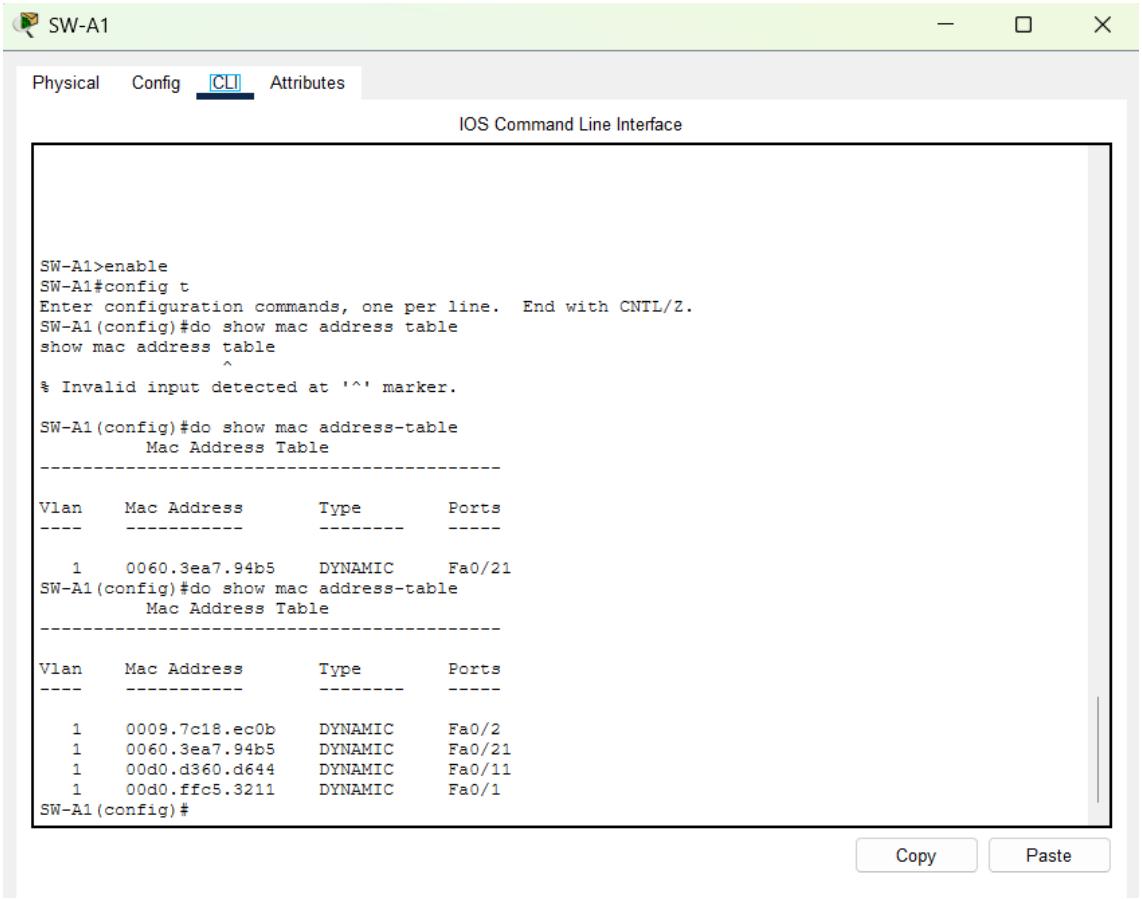
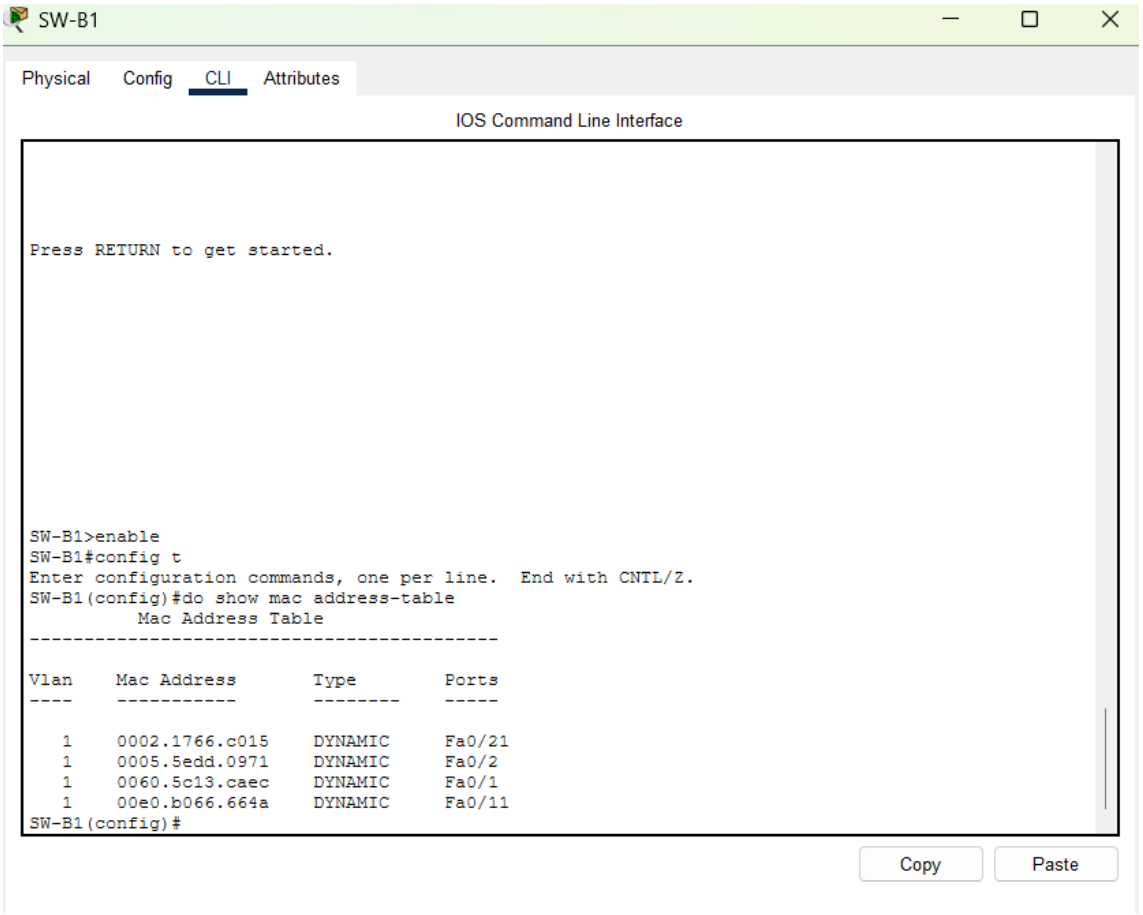


Figure 14

Table of Dynamic MAC Addresses Learned on SW-B1



Note. Created by the author from running *do show mac address-table* on the access switches.

Based on the output obtained in Figure 13, it is confirmed that on switch SW-A1, ports Fa0/1 and Fa0/2 dynamically registered the MAC addresses of workstations PC1-SA (00d0.ffc5.3211) and PC2-SA (0009.7c18.ec0b). Likewise, on SW-B1 the physical addresses of PC1-SB (0060.5c13.caec) were associated with port Fa0/1 and of PC2-SB (0005.5edd.0971) with port Fa0/2. These entries allow each switch to send unicast traffic directly to the corresponding destination device, avoiding unnecessary flooding of the remaining ports.

Likewise, on both switches, interface Fa0/11 registered the physical address of the respective local server: 00d0.d360.d644 for SERVER-A on SW-A1, and 00e0.b066.664a for SERVER-B on SW-B1. This ensures efficient delivery of frames destined for each site's internal services.

Finally, on SW-A1 the dynamic entry on port Fa0/21 (0060.3ea7.94b5) corresponds to interface GigabitEthernet0/0 of router R1_Pi1, while on SW-B1 the entry on Fa0/21 (0002.1766.c015) belongs to interface GigabitEthernet0/0 of R2_Pi2.

When a local device needs to send data outside its own subnet, it uses the ARP protocol to obtain the MAC address of its default gateway. Because this address is registered on port Fa0/21, the switch immediately forwards outbound frames to the LAN interface of the edge router.

4. DHCP

To dynamically provision network parameters to workstations on the currently active segments (test VLAN 10 in access mode), the DHCP service was implemented directly on the Cisco edge routers (R1_Pi1 and R2_Pi2). This configuration ensures the automated assignment of IP addresses, subnet masks, and default gateways, while preserving the ranges required for static addressing (servers, switches, and router interfaces).

Configuring Pools and Exclusions

- Site A (R1_Pi1): The active network is 10.10.0.0/26. The first 9 addresses (10.10.0.1 through 10.10.0.9) were excluded to reserve the router's IP address (10.10.0.1), the management IP address of switch SW-A1 (10.10.0.2), the local server SERVER-A (10.10.0.3), and future critical assignments. The POOL_USUARIOS_SEDE_A pool assigns addresses starting at 10.10.0.10.
- Site B (R2_Pi2): The active network is 10.10.1.0/26. The addresses 10.10.1.1 through 10.10.1.9 were excluded for the router (10.10.1.1), the SW-B1 switch (10.10.1.2), the local server SERVER-B (10.10.1.3), and static infrastructure. The POOL_USUARIOS_SEDE_B pool assigns addresses starting at 10.10.1.10.

Scope of Segmentation

In this prototype, dynamic requests and assignments are strictly limited to active user segments. The subnets corresponding to VLAN 20, VLAN 30, and VLAN 99 remain reserved in the logical design to be associated with their respective DHCP servers once the 802.1Q trunking scheme is implemented in Phase 3.

Network Parameters Assigned to Clients

Table 3

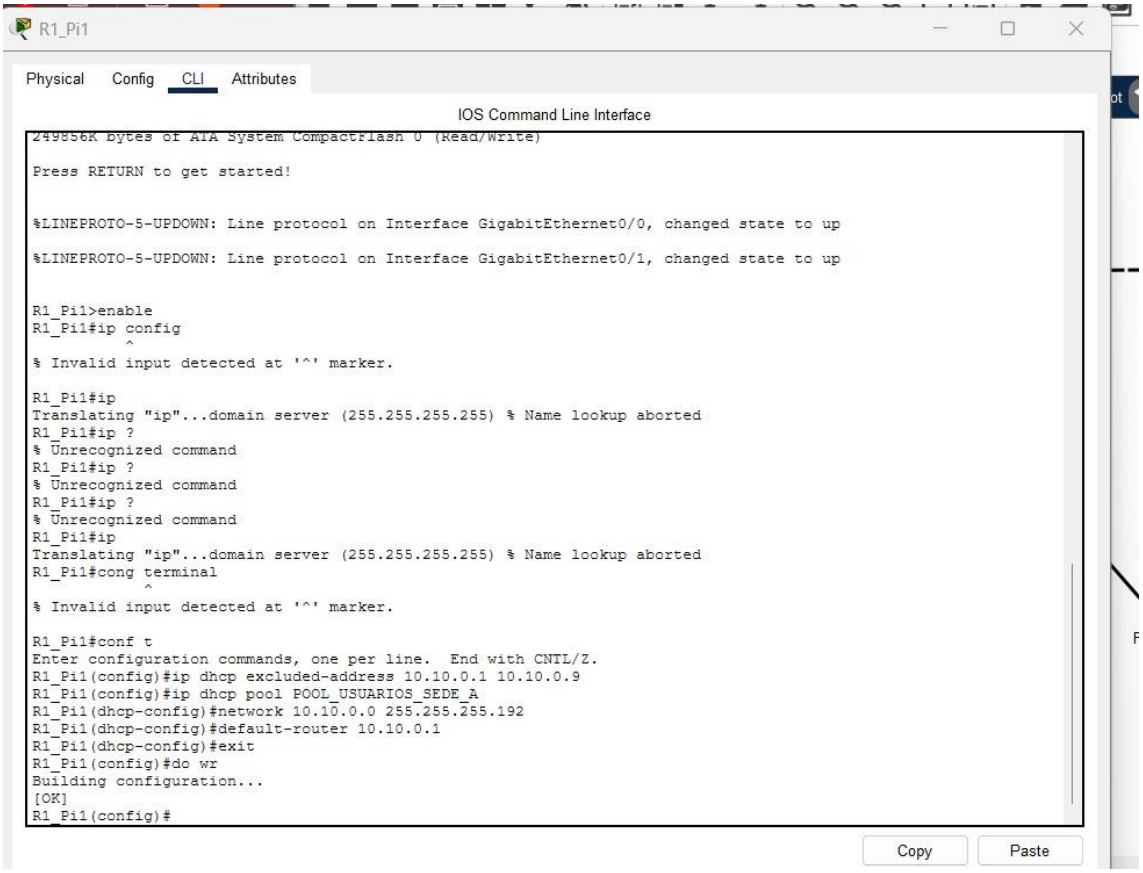
IP Address Settings Assigned by DHCP to Test Clients

Client Device	Headquarters	Segment/Mask	Received IP Address	Gateway	Concession Status
PC1-SA	Site A	10.10.0.0/26 (255.255.255.192)	10.10.0.10	10.10.0.1	Active (Assigned)
PC2-SA	Site A	10.10.0.0/26 (255.255.255.192)	10.10.0.11	10.10.0.1	Active (Assigned)
PC1-SB	Site B	10.10.1.0/26 (255.255.255.192)	10.10.1.10	10.10.1.1	Active (Assigned)
PC2-SB	Site B	10.10.1.0/26 (255.255.255.192)	10.10.1.11	10.10.1.1	Active (Assigned)

Note. Created by the author based on network allocations obtained dynamically in the Cisco Packet Tracer simulation environment.

Figure 15

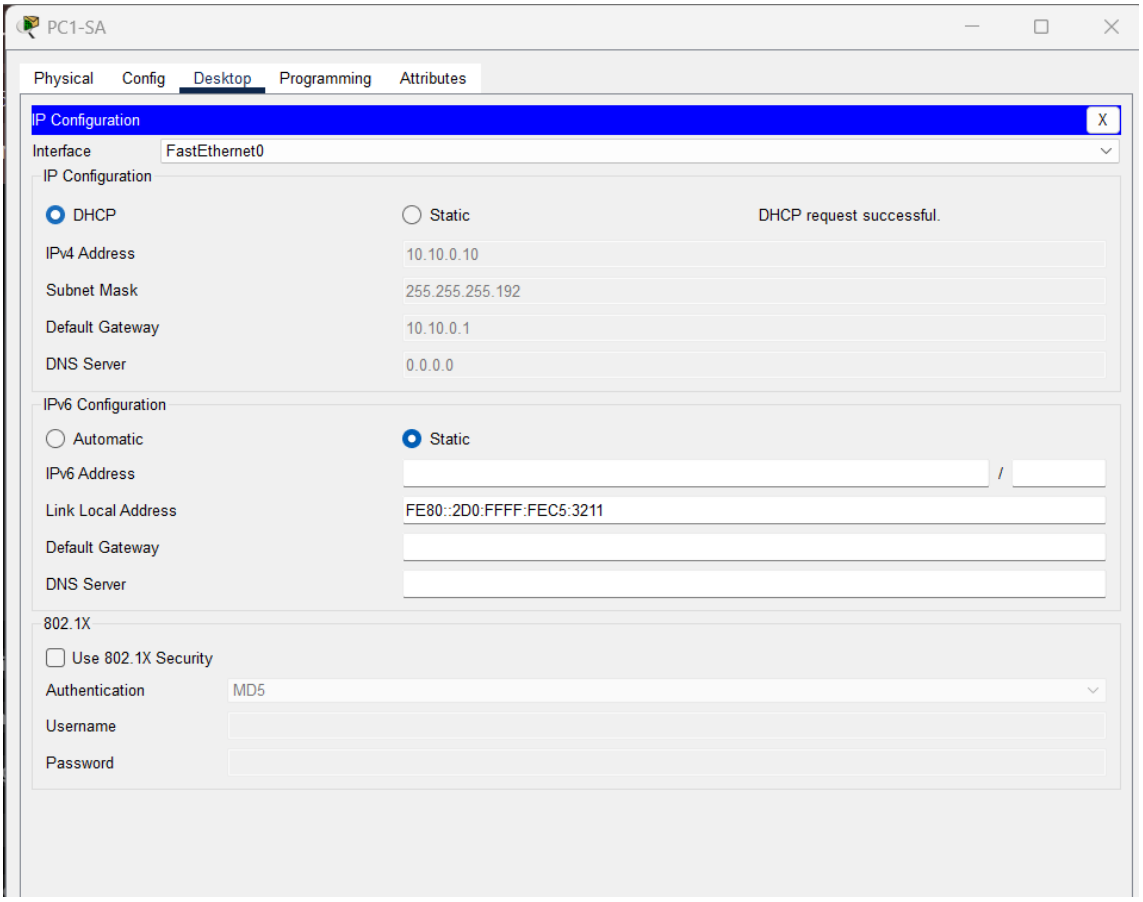
Configuring the DHCP Pool and Exclusions on Router R1_Pi1



Note. Created by the author using the command-line interface (CLI) in Cisco Packet Tracer.

Figure 16

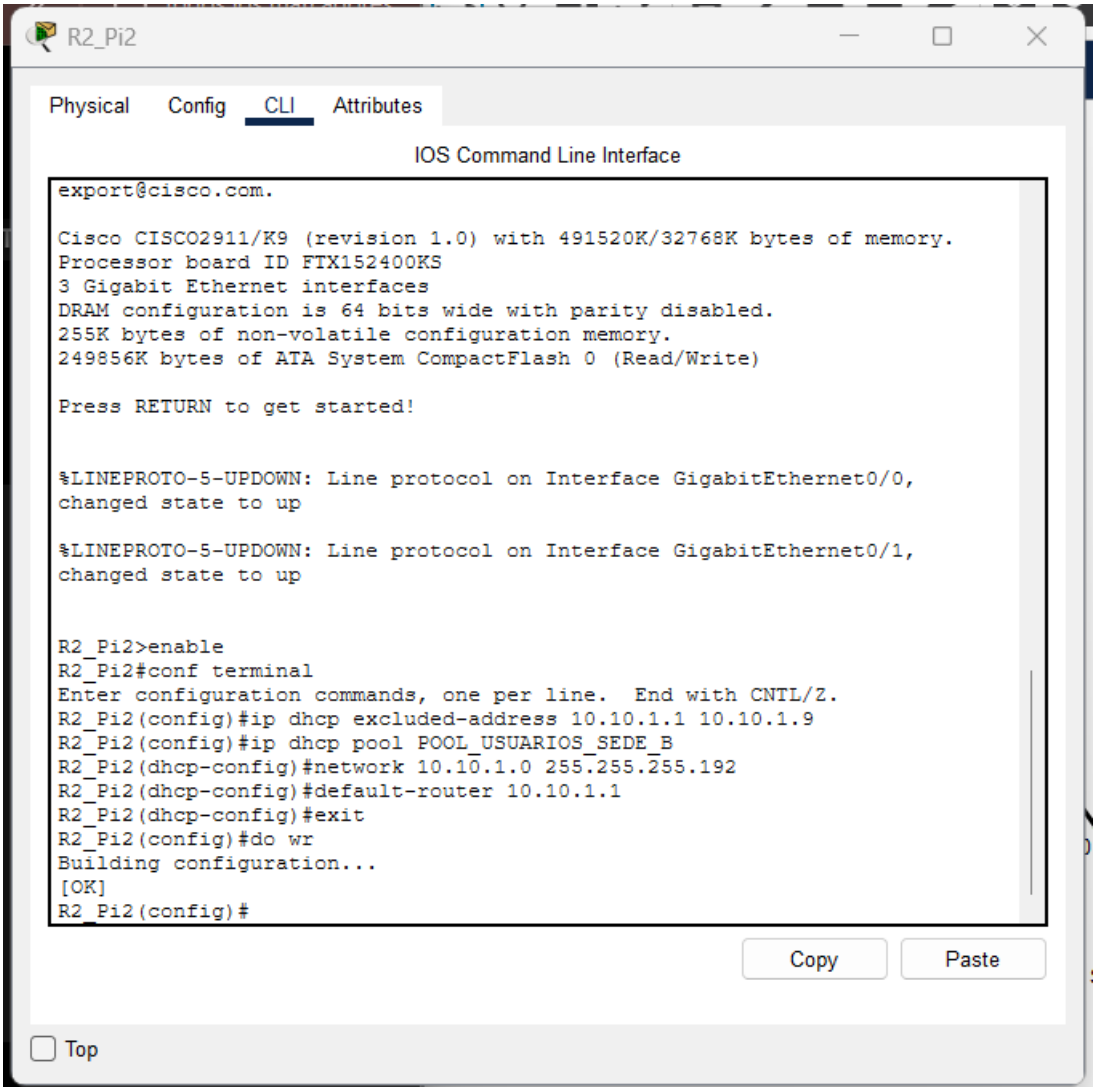
Dynamic IPv4 Parameter Allocation at the PC1-SA Station



Note. Created by the author using the network configuration panel in Cisco Packet Tracer.

Figure 17

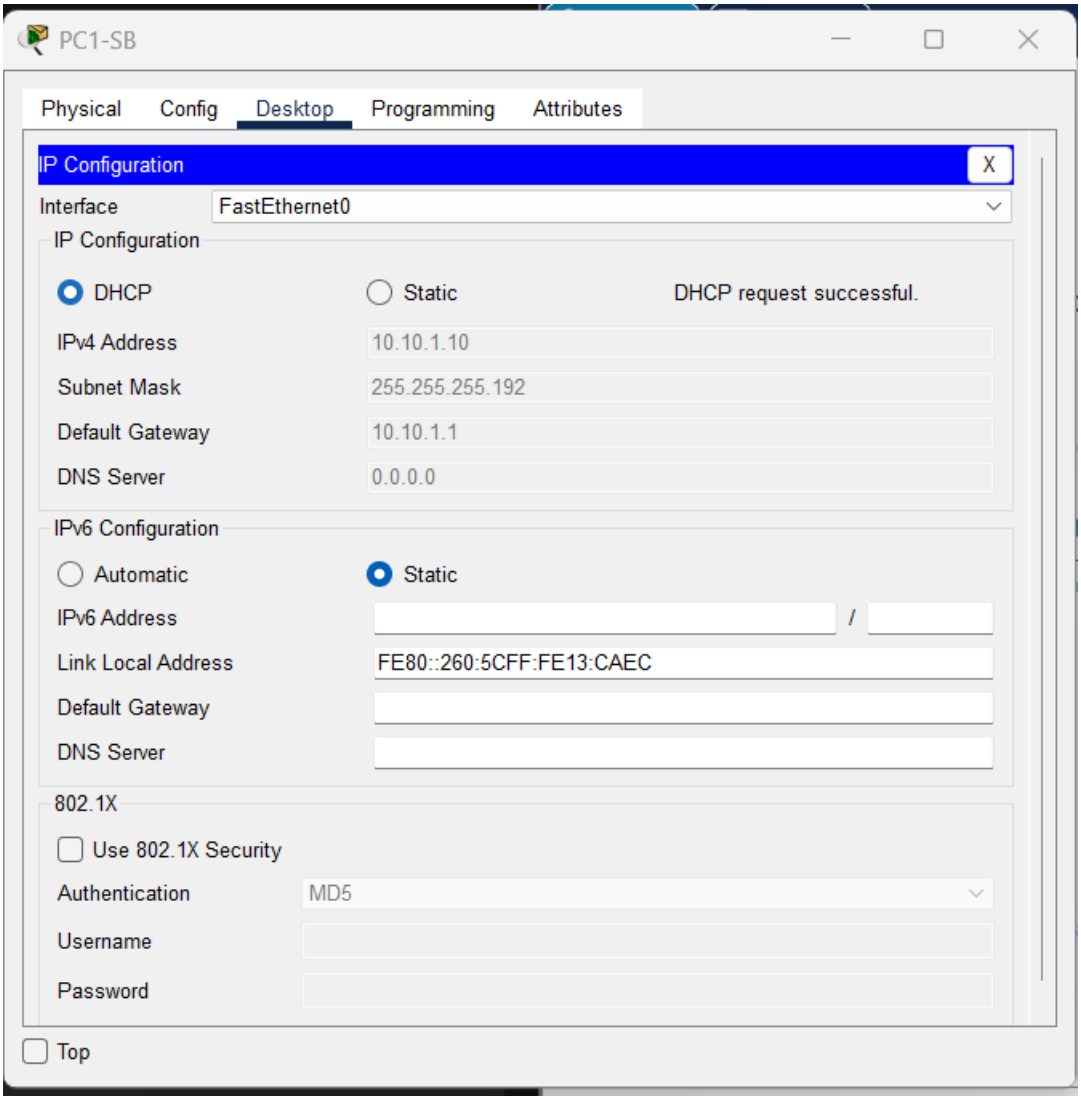
Configuring the DHCP Pool and Exclusions on the R2_Pi2 Router



Nota. Created by the author using the command-line interface (CLI) in Cisco Packet Tracer.

Figure 18

Dynamic Allocation of IPv4 Parameters at PC1-SB Station



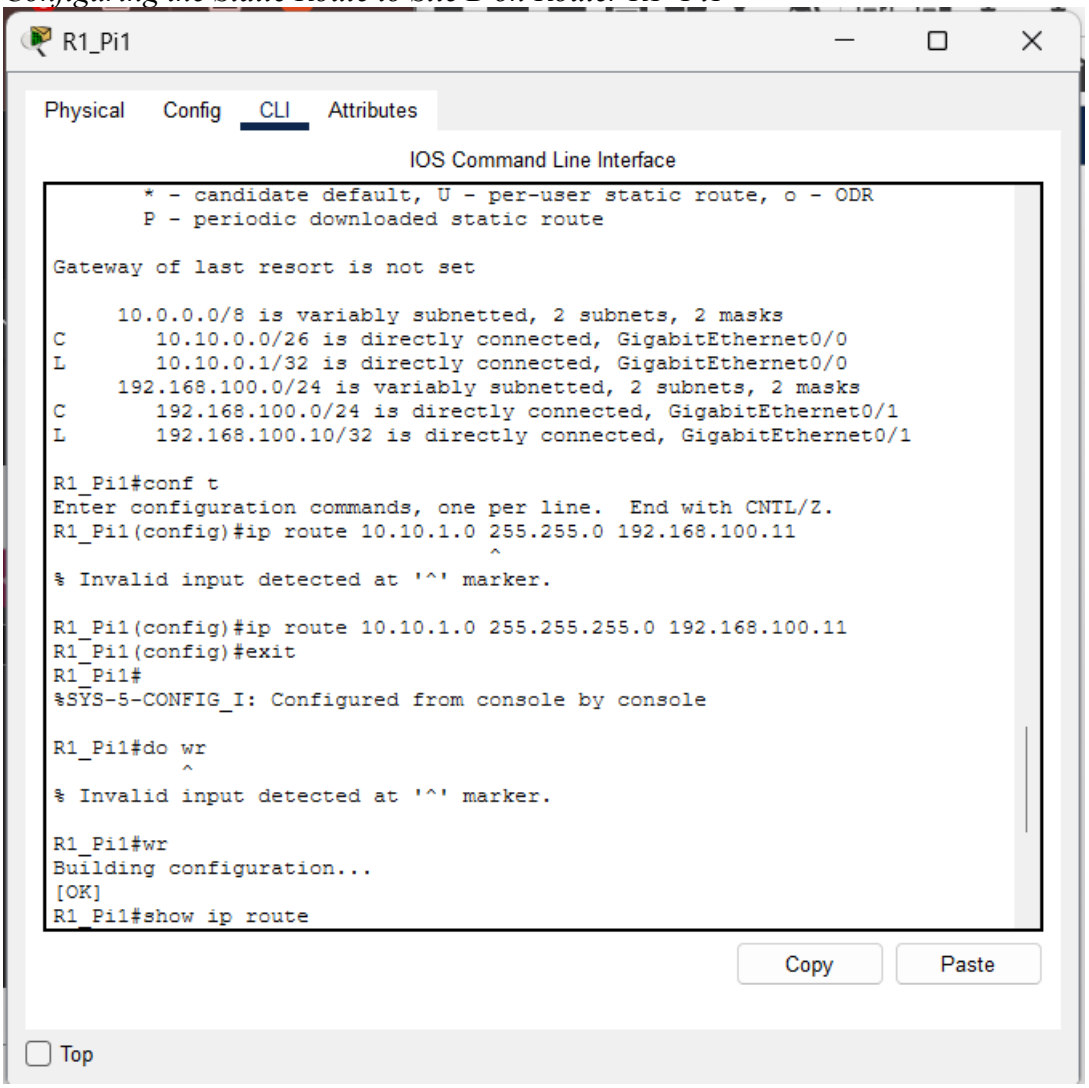
Note. Created by the author using the network configuration panel in Cisco Packet Tracer.

5. INITIAL INTERSITE ROUTING

To connect both sites via the point-to-point WAN link on the 192.168.100.0/24 subnet, temporary static routes were configured on the edge routers R1_Pi1 and R2_Pi2. This configuration ensures direct connectivity between the local subnet at Site A (10.10.0.0/26) and the local subnet at Site B (10.10.1.0/26).

Figure 19

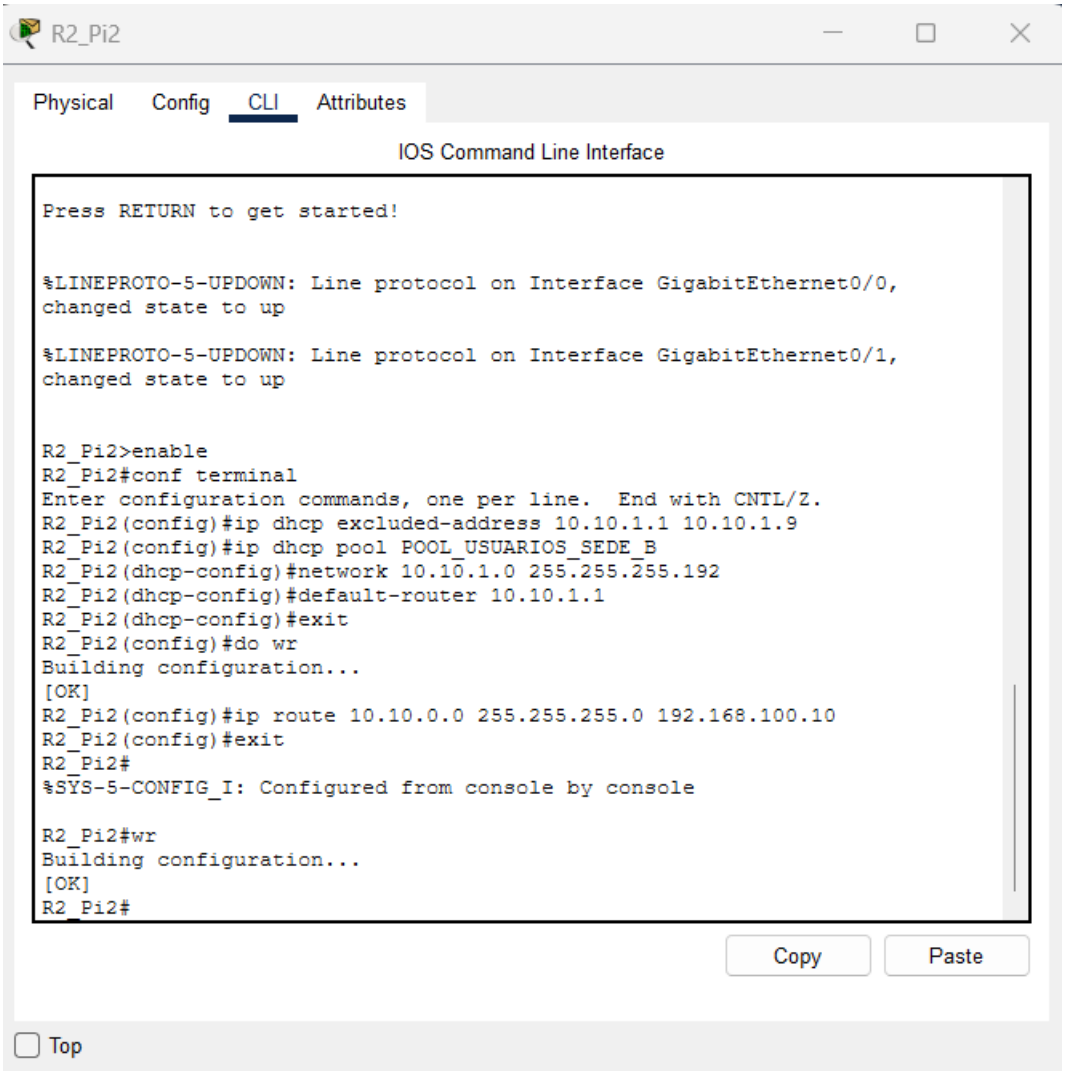
Configuring the Static Route to Site B on Router R1_Pi1



Note. Created by the author in the Cisco Packet Tracer administration console.

Figure 20

Configuring the Static Return Path on the R2_Pi2 Router

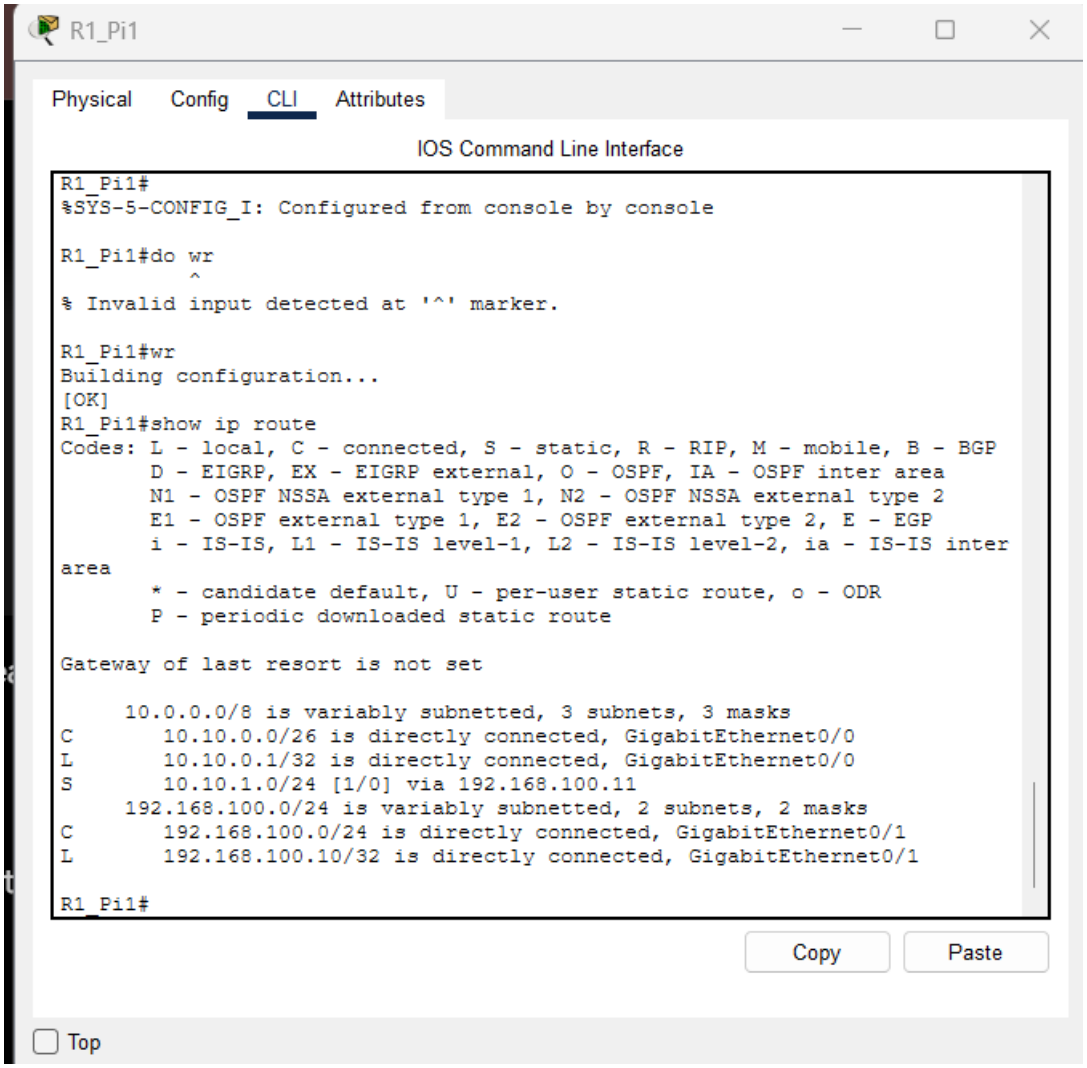


Note. Created by the author in the Cisco Packet Tracer administration console.

As shown in the console output from both routers, the routing tables list the directly connected networks (code C) and the configured static route (code S). On R1_Pi1, traffic destined for the 10.10.1.0/24 network is routed to hop 192.168.100.11 (the WAN interface of R2_Pi2). Conversely, on R2_Pi2, traffic destined for 10.10.0.0/24 is forwarded to hop 192.168.100.10 (the WAN interface of R1_Pi1).

Figure 21

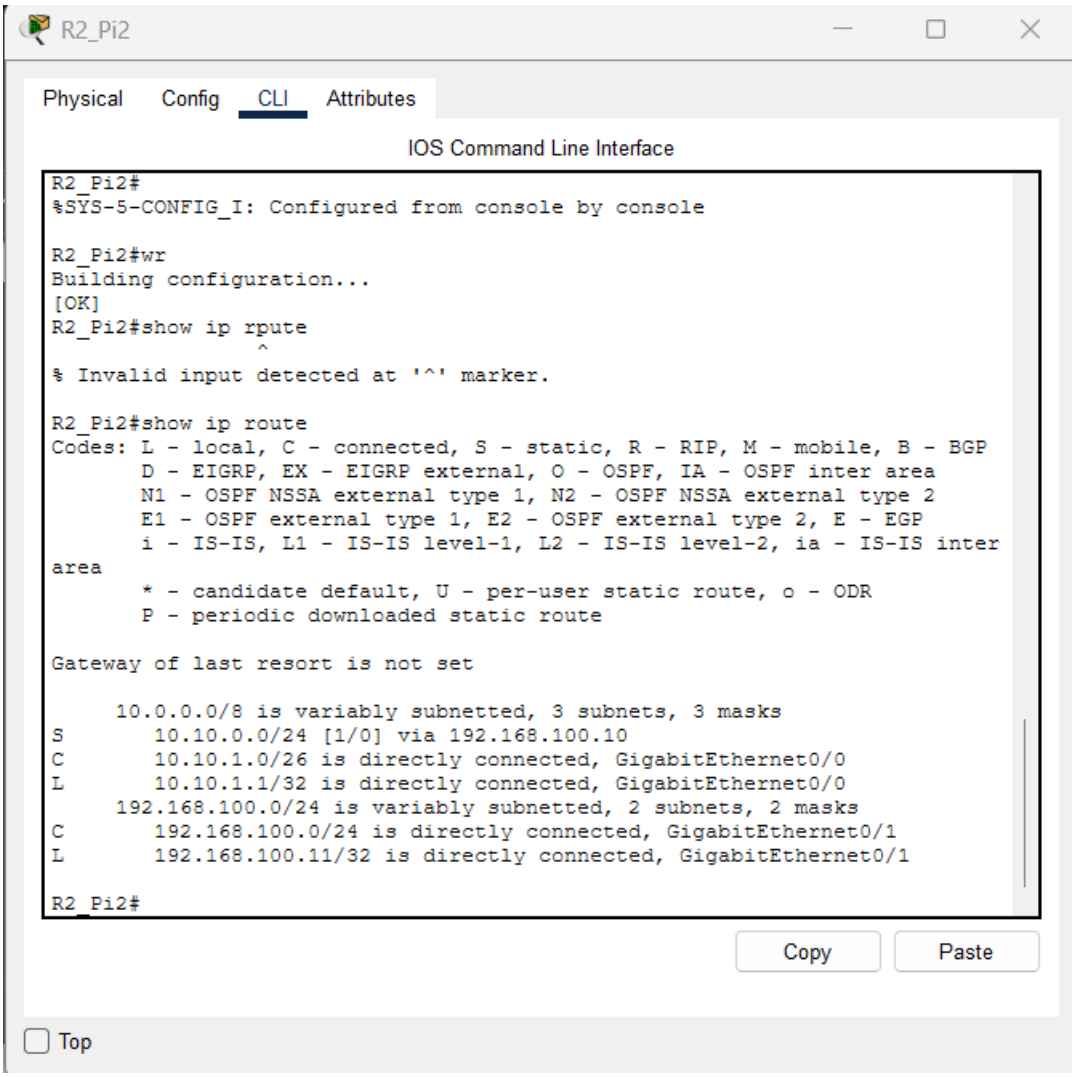
IPv4 Routing Table for the R1_Pi1 Edge Router



Note. Created by the author based on the results of running the "show ip route" command in Cisco Packet Tracer.

Figure 22

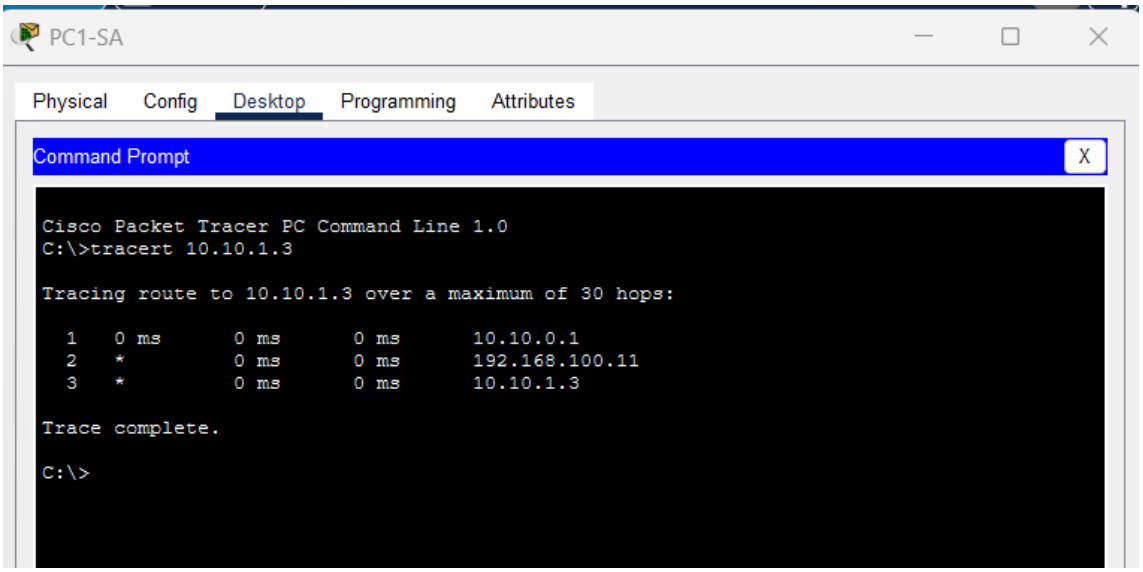
IPv4 Routing Table for the R2_Pi2 Edge Router



Note. Created by the author based on the results of running the “show ip route” command in Cisco Packet Tracer.

Figure 23

Traceability of the Packet Path from PC1-SA to SERVER-B



Note. Created by the author based on the results of running the tracert command in the Cisco Packet Tracer command prompt.

Packet Path Analysis:

The trace test run from the PC1-SA station (10.10.0.10) to the remote server SERVER-B (10.10.1.3) shows the flow across three successive hops:

- Hop 1 (10.10.0.1): The packet leaves the local station and is sent to its default gateway on the GigabitEthernet0/0 interface of router R1_Pi1.
- Hop 2 (192.168.100.11): R1_Pi1 consults its routing table and forwards the traffic over the WAN link to the ingress address of the Branch B router (R2_Pi2). The asterisk at the beginning of this hop corresponds to the ARP resolution time on the inter-site link during the first transmission.
- Hop 3 (10.10.1.3): R2_Pi2 delivers the packet directly to the destination server on Site B's local network via switch SW-B1.

Rationale for Replacing Static Routes with OSPFv2 in Phase 3

Static routes were implemented on a temporary basis in this release to isolate and validate basic Layer 3 connectivity without introducing dynamic convergence variables during initial testing. However, for Phase 3 and deployment on actual hardware (OpenWrt on Raspberry Pi), these routes will be replaced by the OSPFv2 dynamic routing protocol (Area 0) due to:

- Scalability with Full Segmentation: When the planned VLANs (VLAN 20 for Servers, VLAN 30 for Guests, and VLAN 99 for Management) are enabled, OSPF will automatically propagate the new network blocks between sites via Link State Advertisements (LSAs), eliminating the need for manual table updates and reducing the risk of configuration errors.
- Convergence and fault tolerance: OSPF detects interface failures and recalculates paths automatically, improving the availability of the inter-site link.
- Preservation of Local Internet Access: In the OSPF configuration on OpenWrt, the default route will not be advertised, ensuring that each site retains its independent WAN connection to the Internet without public traffic traversing the inter-site link.

6. ADMINISTRATION AND TROUBLESHOOTING

This section details the security guidelines applied for the administrative control of network devices, as well as a technical analysis of an actual incident that occurred during the prototype's implementation in Cisco Packet Tracer.

Secure Administration Policies

In order to mitigate the risk of unauthorized access and comply with the project's guidelines to keep credentials and sensitive information out of the report and the public repository, the following hardening guidelines were applied to the routers and switches:

- System Password Encryption: The global password obfuscation service was enabled using the `service password-encryption` command on all Cisco devices. This prevents terminal or console access passwords from appearing in plain text within the `running-config` file when the device is inspected.

- Cryptographic Protection for Privileged Mode: Access to privileged EXEC mode was configured using the "enable secret" command, which implements a robust hashing algorithm instead of the obsolete "enable password" command.
- Session Control and Inactivity Timers: Session inactivity limits (exec-timeout 5 0) were set on physical console lines (line con 0) and virtual terminal lines (line vty 0 4). This ensures that any open administrative session is automatically closed after 5 minutes of inactivity.
- Isolation in the Final Architecture: In the final design for OpenWrt routers on Raspberry Pi, remote administration will be strictly routed through the secure SSH protocol (TCP port 22), confined to Management VLAN 99, thereby disabling any plaintext access vectors or interfaces exposed to user and guest networks.
- Legal Disclaimers: Legal disclaimers have been configured on the login screen (motd banner), formally stating that unauthorized access to the organization's equipment is prohibited.

Real-World Implementation Case Study: Diagnosis, Correction, and Validation

During the initial configuration of the link between the SW-A1 switch and the R1_Pi1 router, a Layer 3 connectivity issue arose that required a systematic troubleshooting procedure:

- Problem Description: After enabling the DHCP pool on router R1_Pi1, the PC1-SA and PC2-SA stations automatically received their addressing parameters. However, when performing the first verification test using the ping 10.10.0.1 command from the PC1-SA console to its gateway, the test failed, resulting in 100% packet loss (“Request timed out”), preventing traffic from being sent to Site B.
- Diagnosis: A layer-by-layer analysis was performed on the local infrastructure at Site A:
  → Layer 1 and Layer 2 (Physical and Link): The command `show ip interface brief` was executed on R1_Pi1, confirming that the GigabitEthernet0/0

interface was in an operational “up/up” state. Likewise, the link LEDs on switch SW-A1 were green, indicating active synchronization.

→ Forwarding Table (MAC) Inspection: On SW-A1, the `show mac address-table` command was executed. It was found that the switch was not dynamically registering the MAC address of the router’s interface on the trunk/uplink port Fa0/21.

→ Network Parameter Inspection: Upon comparing the `show running-config` output on the router with the clients’ configurations, it was evident that the GigabitEthernet0/0 interface on R1_Pi1 had been configured—due to a typo—with the default /24 subnet mask (10.10.0.1 255.255.255.0), whereas the VLSM plan and the clients were operating under the /26 subnet mask (255.255.255.192). This discrepancy caused inconsistencies in broadcast calculations and in the exchange of ARP frames used to resolve the gateway’s Layer 2 address.

- Corrective Action: We logged into the local network interface of router R1_Pi1 to configure the IP address and its exact subnet mask in accordance with the VLSM design. Additionally, we forced negotiation on the uplink of switch SW-A1:

Adjustment in R1_Pi1:

```
configure terminal
interface GigabitEthernet0/0
ip address 10.10.0.1 255.255.255.192
no shutdown
exit
do write
```

Adjustment in SW-A1:

```
configure terminal
interface FastEthernet0/21
switchport mode access
no shutdown
```

exit

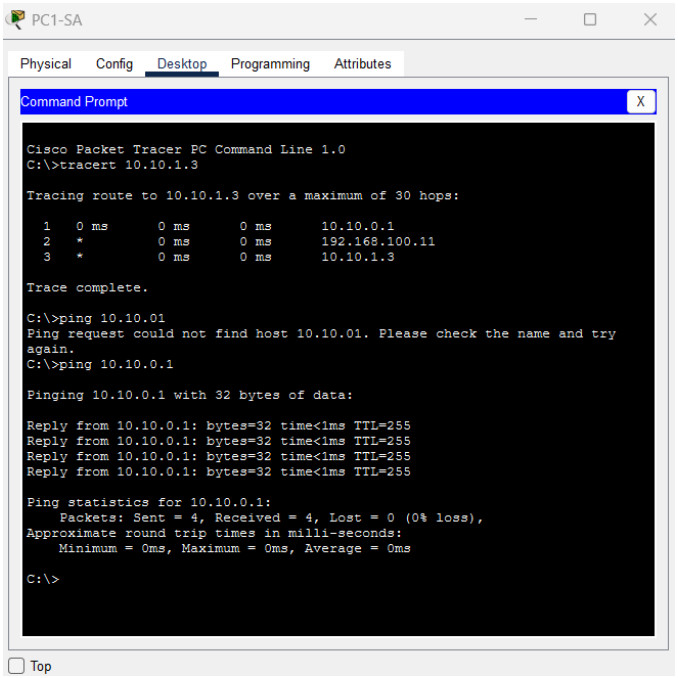
do write

Retesting and Validation:

- From the Command Prompt on PC1-SA, connectivity was refreshed by sending a burst of 4 packets to the corrected gateway address (10.10.0.1).
- The test yielded a 100% success rate (4 packets transmitted, 4 packets received, 0% loss) with response times under 1 ms.
- Finally, the `show mac address-table` command was run again on SW-A1, confirming that the physical address 0060.3ea7.94b5 (belonging to R1_Pi1) was dynamically registered on port Fa0/21, ensuring direct forwarding of outgoing frames.

Figure 24

Successful Connectivity Test to the Gateway Following the Correction in PC1-SA



Note. Created by the author using the Command Prompt in Cisco Packet Tracer. It shows the preliminary route resolution and reachability confirmation to the default gateway 10.10.0.1 with 0% packet loss.

7. PRELIMINARY TESTS

The preliminary tests verify the functions that are active in the Packet Tracer prototype: interface status, local connectivity, MAC learning, DHCP and intersite routing with temporary static routes. Each test has an identifier, a source, a destination, a procedure, an expected result, the observed result, the evidence in this report and a status. A test is marked Passed only when its evidence is shown in this report. Tests for features that are not implemented yet are marked Pending (planned), because they cannot be counted as passed.

Table 4

Preliminary Test Matrix of the Packet Tracer Prototype

ID	Source	Destination	Procedure	Expected result	Observed result	Evidence	Status
PT-01	R1_Pi1, R2_Pi2	Own interfaces	Run do show ip interface brief on both routers.	Gi0/0 and Gi0/1 up/up with the planned addresses.	Gi0/0 and Gi0/1 up/up on both routers: 10.10.0.1 and 192.168.100.10 (R1_Pi1); 10.10.1.1 and 192.168.100.11 (R2_Pi2).	Figure 12	Passed
PT-02	PC1-SA	10.10.0.1 (R1_Pi1)	Run ping 10.10.0.1 (4 packets) from the PC command prompt.	4 replies, 0% loss.	4 sent, 4 received, 0% loss, less than 1 ms.	Figure 24	Passed
PT-03	PC1-SB	10.10.1.1 (R2_Pi2)	Run ping 10.10.1.1 (4 packets) from the PC command prompt.	4 replies, 0% loss.	4 sent, 4 received, 0% loss, less than 1 ms.	Figure 25	Passed
PT-04	SW-A1, SW-B1	Local hosts and router LAN port	Send traffic between end devices and run do show mac address-table on both switches.	Dynamic entries for PCs (Fa0/1, Fa0/2), server (Fa0/11) and router (Fa0/21).	Four dynamic entries on each switch, on ports Fa0/1, Fa0/2, Fa0/11 and Fa0/21.	Figures 13 and 14	Passed
PT-05	PC1-SA	DHCP server on R1_Pi1	Set the NIC to DHCP and wait for the lease.	Address from 10.10.0.10 onward (/26), gateway 10.10.0.1; excluded range not leased.	10.10.0.10, mask 255.255.255.192, gateway 10.10.0.1; “DHCP request successful.” Lease confirmed in the DHCP binding of R1_Pi1.	Figures 16 and 28	Passed
PT-06	PC2-SA	DHCP server on R1_Pi1	Set the NIC to DHCP and wait for the lease.	10.10.0.11, mask 255.255.255.192, gateway 10.10.0.1.	10.10.0.11, mask 255.255.255.192, gateway 10.10.0.1; lease confirmed in	Figures 26 and 28	Passed

					the DHCP binding of R1_Pi1.		
PT-07	PC1-SB	DHCP server on R2_Pi2	Set the NIC to DHCP and wait for the lease.	Address from 10.10.1.10 onward (/26), gateway 10.10.1.1; excluded range not leased.	10.10.1.10, mask 255.255.255.192, gateway 10.10.1.1; “DHCP request successful.” Lease confirmed in the DHCP binding of R2_Pi2.	Figures 18 and 29	Passed
PT-08	PC2-SB	DHCP server on R2_Pi2	Set the NIC to DHCP and wait for the lease.	10.10.1.11, mask 255.255.255.192, gateway 10.10.1.1.	10.10.1.11, mask 255.255.255.192, gateway 10.10.1.1; “DHCP request successful”; lease confirmed in the DHCP binding of R2_Pi2.	Figures 27 and 29	Passed
PT-09	R1_Pi1	Site B network	Run show ip route on R1_Pi1.	Static route (S) to 10.10.1.0/24 via 192.168.100.11 plus connected routes.	S 10.10.1.0/24 [1/0] via 192.168.100.11; connected 10.10.0.0/26 and 192.168.100.0/24.	Figure 21	Passed
PT-10	R2_Pi2	Site A network	Run show ip route on R2_Pi2.	Static route (S) to 10.10.0.0/24 via 192.168.100.10 plus connected routes.	S 10.10.0.0/24 [1/0] via 192.168.100.10; connected 10.10.1.0/26 and 192.168.100.0/24.	Figure 22	Passed
PT-11	PC1-SA	SERVER-B (10.10.1.3)	Run tracert 10.10.1.3 from the PC command prompt.	Three hops: 10.10.0.1, 192.168.100.11, 10.10.1.3.	Hops 10.10.0.1, 192.168.100.11 and 10.10.1.3; asterisks on the first probes (ARP resolution).	Figure 23	Passed
PT-12	PC1-SB	PC1-SA (10.10.0.10)	Run ping 10.10.0.10 (4 packets) from Site B.	4 replies, 0% loss (return path works).	4 sent, 4 received, 0% loss across the WAN link.	Figure 30	Passed
PT-13	R1_Pi1, R2_Pi2	WAN interface of the other router	Run ping 192.168.100.11 from R1_Pi1 and ping 192.168.100.10 from R2_Pi2.	Replies from the WAN interface of Site B.	100 percent success rate (5 of 5 replies) from R1_Pi1 to 192.168.100.11 and from R2_Pi2 to 192.168.100.10.	Figures 31 and 32	Passed
PT-14	R1_Pi1	R2_Pi2	Check OSPFv2 neighbors (Area 0) with show ip ospf neighbor.	Adjacency in FULL state; routes learned by OSPF.	OSPF is not configured in this prototype.	—	Pending (planned)
PT-15	Host in VLAN 10	Host in VLAN 20	Ping across VLANs through Router-on-a-Stick.	Reply only where policy allows it.	VLANs and trunks are not configured yet.	—	Pending (planned)
PT-16	Host in VLAN 10	Router management interface	Attempt SSH from the user network and from VLAN 99.	Denied from VLAN 10; allowed only from VLAN 99.	Management VLAN is not configured yet.	—	Pending (planned)

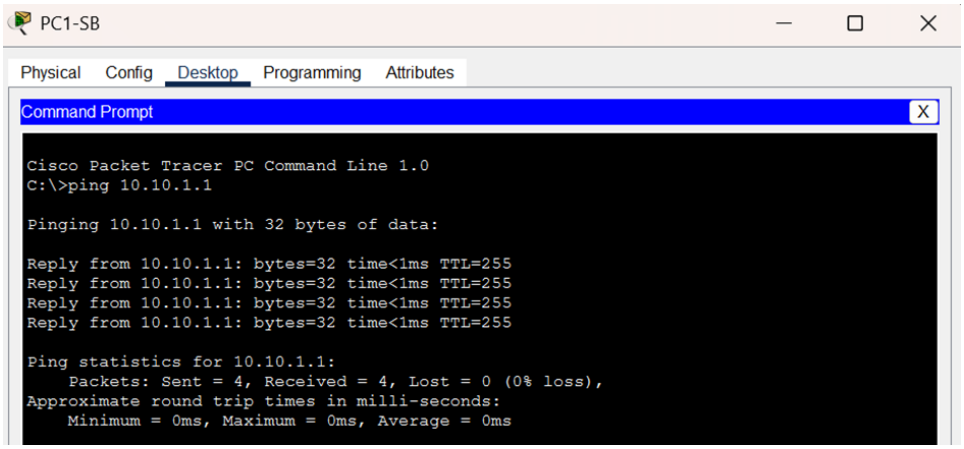
PT-17	Host in VLAN 10	Outside network	Test access to the Internet through NAT.	Translated traffic leaves through the local WAN link.	NAT is not configured yet.	—	Pending (planned)

Note. Created by the author from the tests run in Cisco Packet Tracer. Passed = executed with evidence in this report; Pending (planned) = feature scheduled for Phase 3, not counted as passed.

Of the 17 tests, 13 passed and 4 are pending because the feature is planned for the next phase. All features that are active in the prototype have evidence in this report. The passed tests confirm that the active interfaces are operational, that each site assigns addresses by DHCP, that the switches learn the addresses of the connected devices, and that traffic crosses the WAN link in both directions through the temporary static routes. The evidence for the new tests, which complements Figures 12 to 24, is presented in Figures 25 to 32. Tests PT-14 to PT-17 depend on the VLAN, trunking, OSPF, NAT and firewall work described in Section 8, so they remain pending and are not counted as passed.

Figure 25

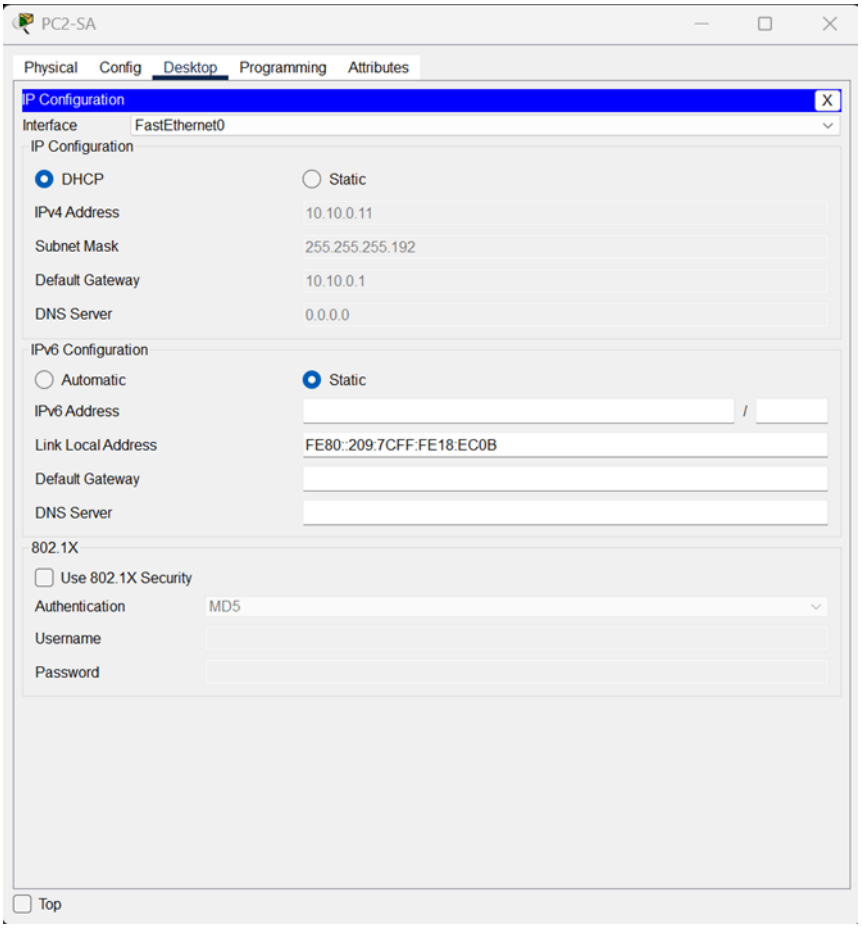
Connectivity Test from PC1-SB to Its Default Gateway



Note. Created by the author using the Command Prompt in Cisco Packet Tracer. It shows 4 packets sent and 4 received to 10.10.1.1 with 0% packet loss.

Figure 26

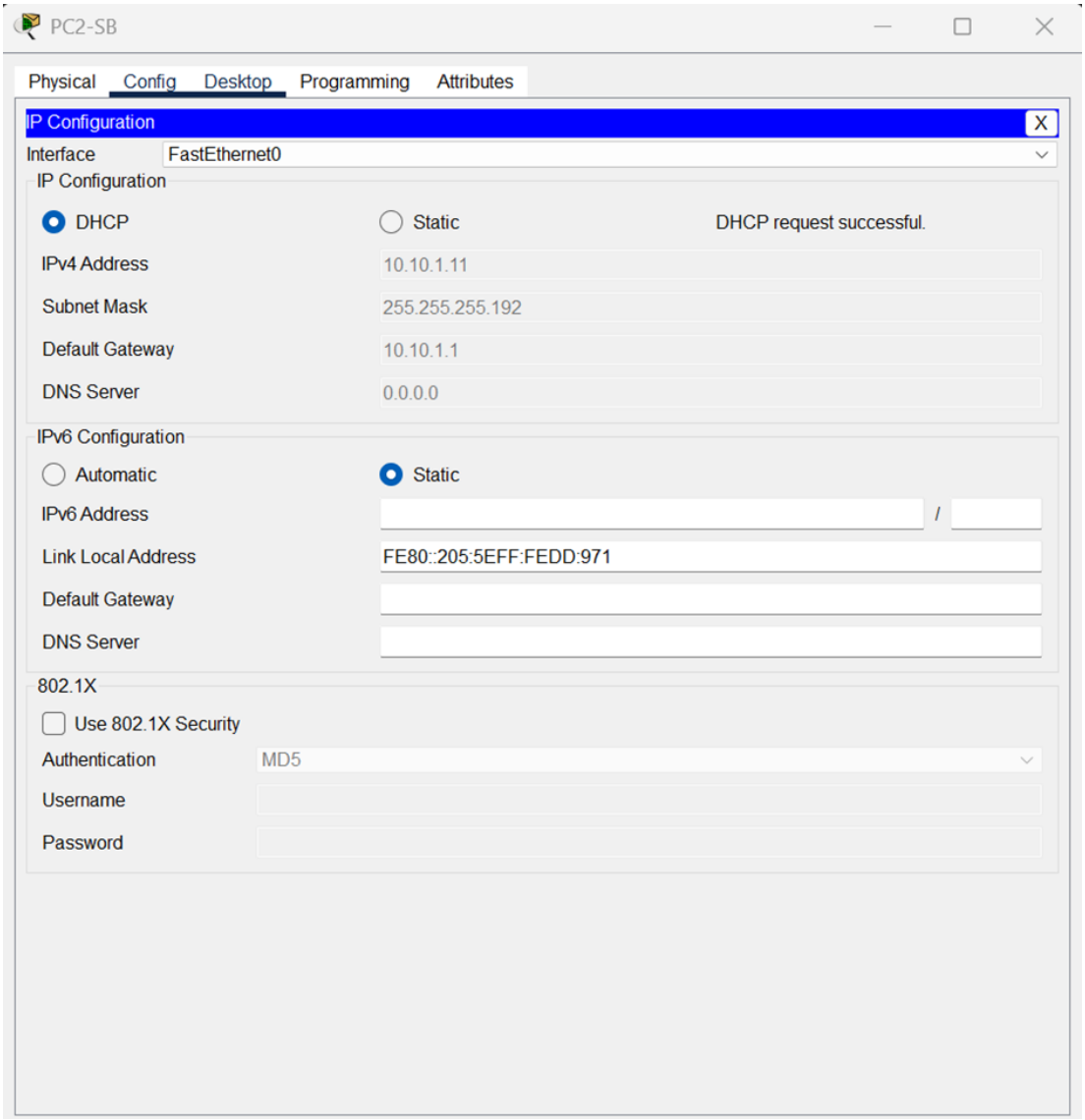
Dynamic IPv4 Parameter Allocation at PC2-SA Station



Note. Created by the author using the network configuration panel in Cisco Packet Tracer.

Figure 27

Dynamic IPv4 Parameter Allocation at PC2-SB Station



Note. Created by the author using the network configuration panel in Cisco Packet Tracer.

Figure 28

DHCP Bindings Registered on the R1_Pi1 Router

```
R1_Pi1>enable
R1_Pi1#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
10.10.0.10       00D0.FFC5.3211          --                      Automatic
10.10.0.11       0009.7C18.EC0B          --                      Automatic
R1_Pi1#
```

Note. Created by the author using the command-line interface (CLI) in Cisco Packet Tracer.

Figure 29

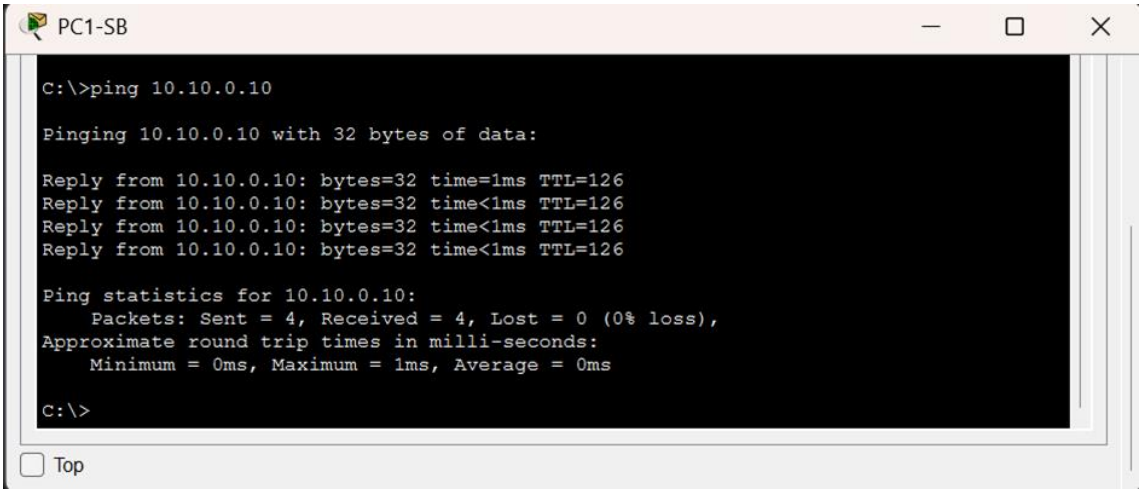
DHCP Bindings Registered on the R2_Pi2 Router

```
R2_Pi2#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
10.10.1.10       0060.5C13.CAEC          --                      Automatic
10.10.1.11       0005.5EDD.0971          --                      Automatic
```

Note. Created by the author using the command-line interface (CLI) in Cisco Packet Tracer.

Figure 30

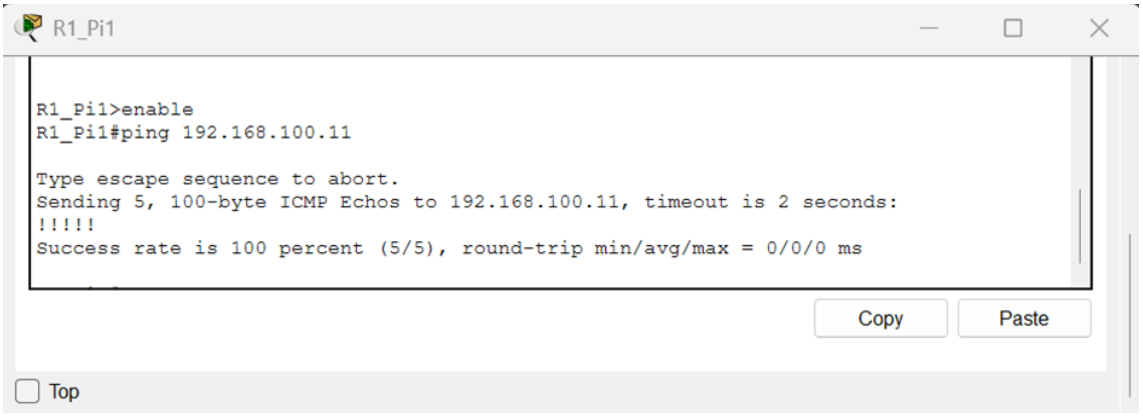
Intersite Connectivity Test from PC1-SB to PC1-SA



Note. Created by the author using the Command Prompt in Cisco Packet Tracer. It shows replies from 10.10.0.10 across the WAN link with 0% packet loss.

Figure 31

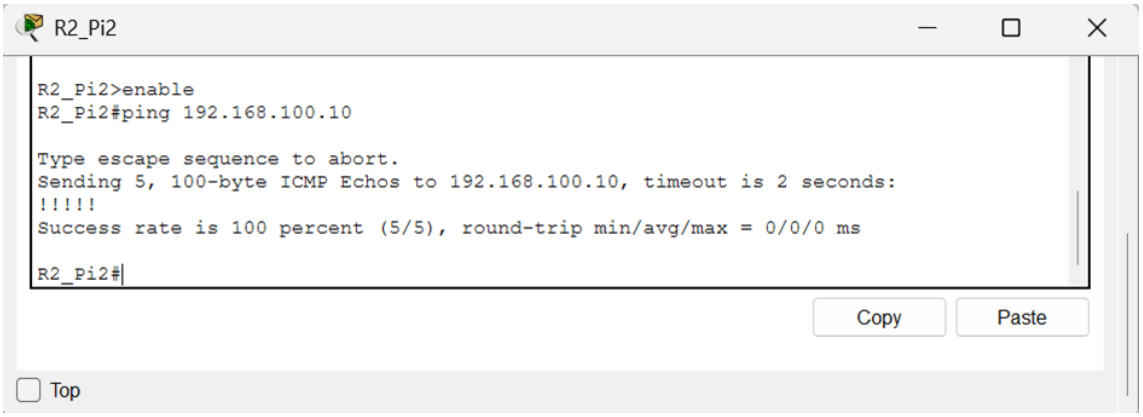
WAN Link Connectivity Test from R1_Pi1 to R2_Pi2



Note. Created by the author using the command-line interface (CLI) in Cisco Packet Tracer. It shows a 100 percent success rate to 192.168.100.11.

Figure 32

WAN Link Connectivity Test from R2_Pi2 to R1_Pi1



Note. Created by the author using the command-line interface (CLI) in Cisco Packet Tracer. It shows a 100 percent success rate to 192.168.100.10.

8. COMPLETION AND MIGRATION PLAN

Remaining Work and Responsibilities

The work not yet implemented is the segmentation, the switching features, dynamic routing and the security services. Table 5 assigns each task to a team member and a target week. The order follows the dependencies between tasks, because the switching content is taught in Week 10.

Table 5

Remaining Work, Responsibilities and Target Weeks

Task	Description	Responsible	Week	Depends on
VLANs 10, 20, 30 and 99	Create the VLANs on both switches and move the access ports to their VLANs.	Santiago Lucina	10	Switching content (Week 10)
802.1Q trunk	Configure the trunk between each switch and its router.	Santiago Lucina	10	VLANs
Router-on-a-Stick	Create one subinterface per VLAN on R1_Pi1 and R2_Pi2 with the gateways of the VLSM plan.	Santiago Lucina	10	Trunk
DHCP per VLAN	Move the user pool to VLAN 10 and add the guest pool for VLAN 30; keep servers and management static.	Santiago Lucina	10–11	Router-on-a-Stick

Second switch and EtherChannel	Add the second switch per site and bundle the redundant links.	Camila Paucar	10	Switching content
RSTP	Enable rapid spanning tree and choose the root bridge to prevent loops.	Camila Paucar	10	Second switch
OSPFv2	Replace the static routes with OSPF Area 0 between sites; do not advertise a default route.	Camila Paucar	11	VLAN subnets
NAT and firewall	Configure NAT on the WAN interface and rules that separate users, guests, servers and management.	Pamela Toapanta	11	VLANs and OSPF
Secure administration	Restrict SSH to VLAN 99 and repeat the hardening already applied (secret, timeouts, banner).	Pamela Toapanta	11	VLAN 99
Complete retest	Repeat PT-01 to PT-17, add the evidence for the new features and record the final status.	Pamela Toapanta	11	All of the above

Note. Created by the author. Responsibilities were distributed so that each member leads a block of the remaining configuration.

Migration from Cisco to OpenWrt

The Cisco configuration cannot be copied into OpenWrt because the two systems use different configuration models. In Packet Tracer each function is configured with CLI commands, while OpenWrt stores its configuration in text files under /etc/config and applies it by service. What carries over from the prototype is the design: addresses, masks, VLAN identifiers, DHCP ranges, routes and security rules. Table 6 shows how each concept of the prototype is expressed in OpenWrt.

Table 6
Translation of Cisco Configuration Concepts to OpenWrt

Cisco concept (Packet Tracer)	OpenWrt equivalent	Remark
Interface addressing (ip address)	Interface sections in /etc/config/network, with the address and mask of each network.	The values come from the VLSM plan; the syntax is different.
VLANs and 802.1Q subinterfaces (Router-on-a-Stick)	VLAN-tagged devices over the physical port, with one OpenWrt interface per VLAN, also defined in /etc/config/network.	A Raspberry Pi has few ports, so the tagged link goes to the Cisco switch trunk.
DHCP pools and excluded addresses	DHCP service (dnsmasq) configured in /etc/config/dhcp, with the start and size of the range and static leases.	The excluded range is obtained by starting the range after the reserved addresses.

Static routes (ip route)	Route sections in /etc/config/network, with target, mask and gateway.	Same idea as in Packet Tracer; used only until OSPF is working.
OSPFv2	Not part of the default OpenWrt installation; installed as an extra routing package (FRRouting or BIRD).	Area 0, networks and no default-route advertisement are kept as design decisions.
NAT (ip nat inside/outside)	Masquerading enabled on the WAN zone in the firewall configuration (/etc/config/firewall).	Each site keeps its own Internet access.
Access lists and security between VLANs	Firewall zones, forwarding rules and traffic rules in /etc/config/firewall.	Rules are written per zone instead of per interface.
EtherChannel and RSTP	Stay on the Cisco switches; the OpenWrt router only connects to the switch trunk.	Link aggregation on the Raspberry Pi would need extra network adapters.
Secure administration (secret, SSH, timeouts)	SSH server (Dropbear) listening only on the management interface, key-based login and a firewall rule for VLAN 99.	Credentials are not stored in the report or the repository.
Configuration backup (write memory)	Backup of the /etc/config files and of the router configuration archive.	Saved in the repository without passwords or keys.

Note. Created by the author. The Cisco switches keep their own configuration; only the routers are replaced by OpenWrt equipment.

The migration will follow this sequence: (a) save the Cisco configurations and the addressing table as the reference; (b) install OpenWrt and name the devices; (c) configure interfaces and addressing; (d) configure DHCP; (e) add the static route and test with the same procedures as the matrix; (f) add the VLANs over the trunk; (g) install and configure OSPF; (h) configure NAT and the firewall; (i) restrict SSH to VLAN 99; (j) repeat the preliminary tests, keeping the IDs of Table 4 and adding the hardware results.

Updated Schedule

Table 7

Schedule for the Complete Simulation and the Initial Hardware Deployment

Week	Milestone
Week 9	Deliver the Packet Tracer prototype, this report and the test matrix.
Week 10	Equipment distribution. Configure VLANs, trunks, Router-on-a-Stick, EtherChannel and RSTP in the simulation.
Week 11	Replace static routes with OSPF, add NAT, firewall and secure administration in the simulation, and complete the retest. Start the initial hardware deployment: install OpenWrt, configure interfaces, DHCP and the first route.

Week 15	Final review of the configuration and the documentation.
Week 16	Final presentation.

Note. Created by the author. The schedule follows the project continuity defined for the course.